JEE-Advanced-28-08-2022 [Paper-1]

Physics

Question: Two spherical stars A and B have densities ρ_A and ρ_B, respectively. A and B have the same radius, and their masses M_A and M_B are related by $M_B = 2M_A$. Due to an interaction process, star A loses some of its mass, so that its radius is halved, while its spherical shape is retained and its density remains ρ_A. The entire mass lost by A is deposited as a thick spheric shell on B with the density the shell being ρ_A. If v_A and v_B are the escape velocities from A and B after the interaction process, the ratio $\frac{v_B}{v_A} = \sqrt{\frac{10n}{15^{1/3}}}$. The value of n is ________

Answer: 2.30

Solution:

$$v_A = \sqrt{\frac{2GM_A}{8\times\left(\frac{R}{A}\right)}} = \frac{v_0}{2}$$

For, B, $\frac{4}{3}\pi\left(r^3 - R^3\right) = \frac{4}{3}\pi R^3 \times \frac{7}{8}$

$$\Rightarrow r = \left(\frac{15}{8}\right)^{\frac{1}{3}} R$$

Therefore, $v_B = \sqrt{\frac{2G\times\left(2M_A + \frac{7}{8}M_A\right)}{\frac{(15)^{\frac{1}{3}}R}{2}}} = v_0 \times \sqrt{\frac{23\times 2}{8\times(15)^{\frac{1}{3}}}}$

Therefore, $\frac{v_B}{v_A} = \sqrt{\frac{23}{(15)^{\frac{1}{3}}}} = \sqrt{\frac{2.30\times 10}{(15)^{\frac{1}{3}}}}$

Therefore, n = 2.30

Question: The minimum kinetic energy needed by an alpha particle to cause the nuclear reaction ${}^{16}_{7}N + {}^{4}_{2}He \rightarrow {}^{1}_{1}H + {}^{19}_{8}O$ in a laboratory frame is n (in MeV). Assume that ${}^{16}_{7}N$ is at rest in the laboratory frame. The masses of ${}^{16}_{7}N$, ${}^{1}_{1}H$ and ${}^{19}_{8}O$ can be taken to be 16.006 u, 4.003 u, 1008 u and 19.003 u, respectively, where 1 u = 930 MeV c^{-2}. The value of n is _______.

Answer: 2.33

Solution:

$Q = (m_N + m_{He} - m_H - m_O) \times c^2$

$= (16.006 + 4.003 - 1.008 - 19.003) \times 930 MeV$

$= -1.86 MeV$

$= 1.86 MeV$ energy absorbed.

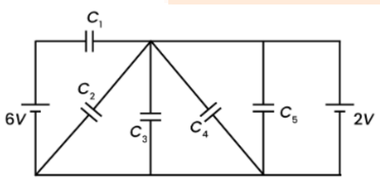
And, $\frac{1}{2} \times \frac{m \times 4m}{5m} \times v^2$ = max loss in kinetic energy

$\Rightarrow \frac{1}{2} mv^2 = \frac{5}{4} \times Q$

$= \frac{5}{4} \times (1.86) MeV = 2.325 MeV$

Therefore, $n = 2.33$

Question: In the following circuit $C_1 = 12\mu F, C_2 = C_3 = 4\mu F$ and $C_4 = C_5 = 2\mu F$. The charge stored in C_3 is _______μC



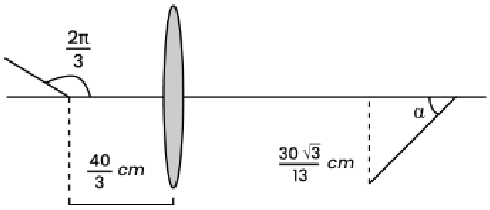
Answer: 8.00

Solution:

From circuit given, Potential difference across C_3 is 2 V (constant)

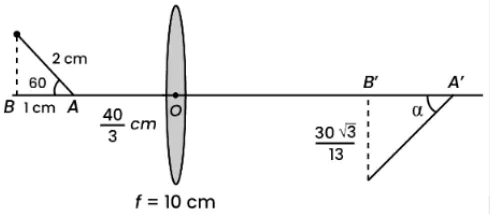
Therefore, $Q_3 = 2 \times 4\mu C = 8\mu C$

Question: A rod of length 2 cm makes an angle $\frac{2\pi}{3}$ rad with the principal axis of a thin convex lens. The lens has a focal length of 10 cm and placed at a distance of $\frac{40}{3} cm$ from the object as shown in the figure. The height of the image is $\frac{30\sqrt{3}}{13}$ cm and the angle made by it with respect to the principal axis is α rad. The value of α is $\frac{\pi}{n}$ rad, where n is _______



Answer: 6.00

Solution:



$$OA' = \frac{\frac{40}{3}\times 10}{\frac{43}{3}-10} = 40cm$$

$$OB' = \frac{\frac{43}{3}\times 10}{\frac{43}{3}-10} = \frac{430}{13}cm$$

Therefore, $A'B' = 40 - \frac{430}{13} = \frac{90}{13}cm$

Therefore, $\tan\alpha = \frac{30\sqrt{3}}{13\left(\frac{90}{13}\right)} = \frac{1}{\sqrt{3}}$

$$\Rightarrow \alpha = \frac{\pi}{6}$$

Therefore, $n = 6.00$

Question: At time t = 0, a disk of radius 1 m starts to roll without slipping on a horizontal plane with an angular acceleration of $\alpha = \frac{2}{3} rad\, s^{-2}$. A small stone A stuck to the disk. At t = 0, it is at the contact point of the disk and the plane. Later, at time $t = \sqrt{\pi} s$, the stone detaches itself and flies off tangentially from the disk. The maximum height (in m) reached by the stone measured from the plane is $\frac{1}{2}+\frac{x}{10}$. The value of x is ____________ [Take g = 10 ms^{-2}]

Answer: 0.52

Solution:

The angle rotated by disc in $t=\sqrt{\pi}s$ is

$$\theta=\omega_0 t+\frac{1}{2}\alpha t^2$$

$$\Rightarrow \theta=\frac{1}{2}\times\frac{2}{3}\left(\sqrt{\pi}\right)^2$$

$$=\frac{\pi}{3}rad$$

and the angular velocity of disc is

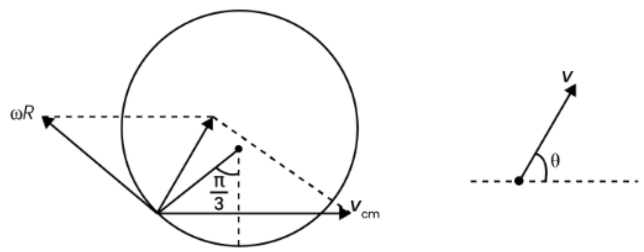
$$\omega=\omega_0+\alpha t$$

$$=\frac{2\sqrt{\pi}}{3}rad/s$$

And $v_{cm}=\omega R=\frac{2\sqrt{\pi}}{3}\times 1$

$$=\frac{2\sqrt{\pi}}{3}m/s$$

So, at the moment it detaches the situation is



$$v=\sqrt{\left(\omega R\right)^2+v_{cm}^2+2\left(\omega R\right)v_{cm}\cos 120°}$$

$$=v_{cm}=\frac{2\sqrt{\pi}}{3}m/s$$

and $\tan\theta=\dfrac{\omega R\sin 120°}{v_{cm}+\omega R\cos 120°}$

$\Rightarrow \tan\theta = \sqrt{3}$

$\Rightarrow \theta = \frac{\pi}{3} rad$

So, $H_{\max} = \frac{u^2 \sin^2\theta}{2g}$

$= \frac{\left(\frac{2\sqrt{\pi}}{3}\right)^2 \times \sin^2 60°}{2\times 10}$

$= \frac{4\pi \times 3}{9\times 2\times 10\times 4}$

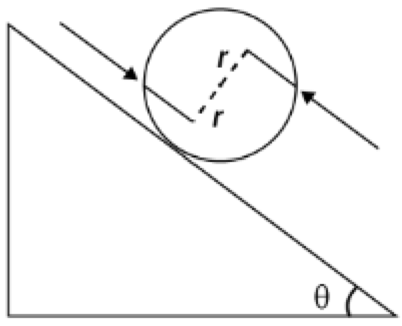
$= \frac{\pi}{60} m$

So, height from ground will be

$R(1-\cos 60°) + \frac{\pi}{60} = \frac{1}{2} + \frac{x}{10}$

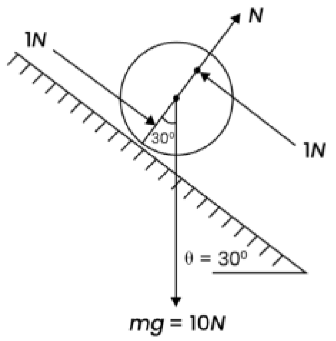
$\Rightarrow x = \frac{\pi}{6} = 0.52$

Question: A solid sphere of mass I kg and radius 1 m rolls without slipping on a fixed inclined plane with an angle of inclination $\theta = 30^{o}$ from the horizontal. Two forces of magnitude 1 N each, parallel to the incline, act on the sphere, both at distance r = 0.5 m from the center of the sphere, as shown in the figure. The acceleration of the sphere down the plane is _____ms^{-2} (Take g = 10 ms^{-2})



Answer: 0.286

Solution:



Taking torque about contact point

$\vec{\tau} = mgR\sin 30^{\otimes} + 1\times 1^{\odot}$

taking $\otimes$ is positive

$5 - 1 = \frac{7}{5}mR^2\alpha$

$\Rightarrow \alpha = \frac{20}{7} rad/s^2$

So, $a_{cm} = \alpha R = \frac{20}{7} m/s^2$

$\boxed{a_{cm} = 2.86\, m/s^2}$

Question: Consider an LC circuit, with inductance L = 0.1 H and capacitance C = 10^{-3}F, kept on a plane. The area of the circuit is 1 m^2. It is placed in a constant magnetic field of strength B_o which is perpendicular to the plane of the circuit. Al time t = 0. the magnetic field strength starts increasing linearly as $B = B_0 + \beta t$ with $\beta = 0.04 Ts^{-1}$. The maximum magnitude of the current in the circuit is ___________ mA.

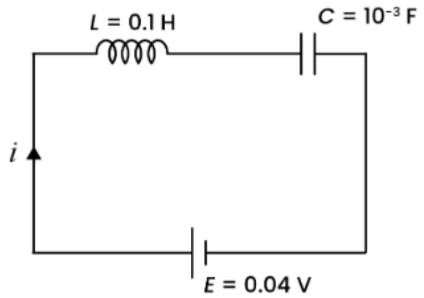
Answer: 04.00

Solution:

Emf inducted in the circuit is

$$|E| = \left|\frac{d\phi}{dt}\right| = \frac{d}{dt}\left(\left(B_0 + \beta t\right)A\right)$$
$$= \beta \times A$$
$$= 0.04V$$

So the circuit can be rearranged as



Using Kirchhoff's law we can write

$$E = L\frac{di}{dt} + \frac{q}{C}$$

$$L\frac{di}{dt} = E - \frac{q}{C}$$

$$or\,\frac{d^2q}{dt^2} = -\frac{1}{LC}(q - CE)$$

Using SHM concept we can write

$$q = CE + A\sin(\omega t + \phi)\left(\text{where}\,\omega = \frac{1}{\sqrt{LC}}\right)$$

at $t = 0, q = 0 \,\&\, i = 0$

So $A = CE \,\&\, \phi = -\frac{\pi}{2}$

$$q = CE - CE\cos\omega t$$

$$\text{so}\, i = \frac{dq}{dt} = CE\omega\sin\omega t$$

$$\text{so}\, i_{\max} = \frac{10^{-3} \times 0.04}{\sqrt{0.1 \times 10^{-3}}}$$

=4 mA

Question: A projectile is fired from horizontal ground with speed v and projection angle θ. When the acceleration due to gravity is g, the range of the projectile is d. If at the highest point in its trajectory, the projectile enters a different region where the effective acceleration due to gravity is $g' = \frac{g}{0.81}$, then the new range is d' = nd. The value of n is _____

Answer: 00.95

Solution:

$$d=\frac{u^2\sin 2\theta}{g}$$
$$H=\frac{u^2\sin^2\theta}{2g}$$

So, after entering in the new region, time taken by projectile to reach ground

$$t=\sqrt{\frac{2H}{g'}}$$
$$=\sqrt{\frac{2u^2\sin^2\theta\times 0.81}{2g\times g}}$$
$$=\frac{0.94\sin\theta}{g}$$

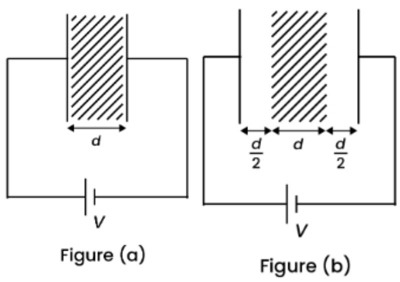
So, horizontal displacement done by the projectile in new region is

$$x=\frac{0.9u\sin\theta}{g}\times u\cos\theta$$
$$=0.9\frac{u^2\sin 2\theta}{2g}$$
$$So, d'=\frac{d}{2}+x$$

= 0.95 d

So, n = 0.95d

Question: A medium having dielectric constant K >1 fills the space between the plates of a parallel plate capacitor. The plates have large area, and the distance between them is d. The capacitor is connected to a battery of voltage V, as shown in Figure (a). Now, both the plates are moved by a distance of $\frac{d}{2}$ from their original positions, as shown in Figure (b)



In the process of going from the configuration depicted in Figure (a) to that in Figure (b), which of the following statement(s) is(are) correct?

Options:

(a) The electric field inside the dielectric material is reduced by a factor of $2K$.

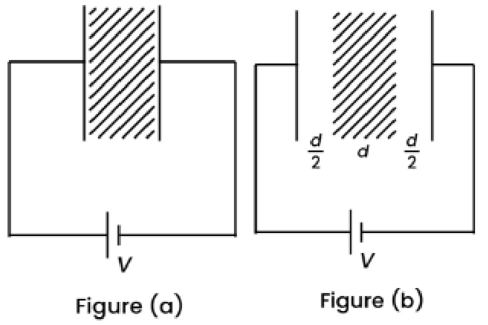
(b) The capacitance is decreased by a factor of $\frac{1}{K+1}$.

(c) The voltage between the capacitor plates is increased by a factor of $(K + 1)$.

(d) The work done in the process DOES NOT depend on the presence of the dielectric material.

Answer: (b)

Solution:



$$C_a = \frac{K\varepsilon_0 A}{d}$$

$$q_a = \frac{K\varepsilon_0 A}{d}V$$

$$E_a = \frac{q_a}{KA\varepsilon_0} = \frac{K\varepsilon_0 AV}{dK\varepsilon_0 A}$$

$$= \frac{V}{d}$$

$$C_b = \frac{\varepsilon_0 A}{d + \left(\frac{d}{K}\right)}$$

$$= \frac{\varepsilon_0 AK}{d(K+1)}$$

$$q_b = \frac{\varepsilon_0 AKV}{d(K+1)}$$

$$(E_b)_{dielectric} = \frac{E_{air}}{K}$$

$$= \frac{q_b}{KA\varepsilon_0}$$

$$= \frac{\varepsilon_0 AKV}{d(K+1)(KA\varepsilon_0)}$$

$$= \frac{V}{d(K+1)}$$

Capacitance decrease by a factor of $\frac{1}{K+1}$

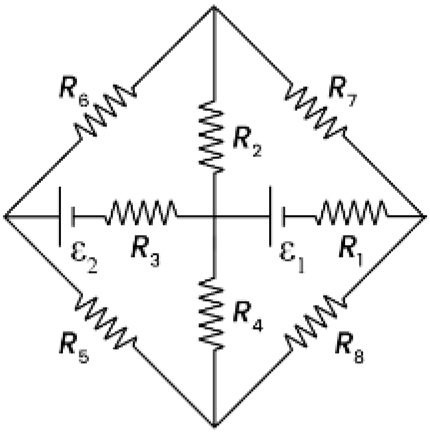
Work done in the process = $U_f - U_i$

$$\frac{1}{2}(C_f - C_i)V^2$$

$$= \frac{1}{2}\left(\frac{\varepsilon_0 AK}{d(K+1)} - \frac{K\varepsilon_0 A}{d}\right)V^2$$

$$= \frac{1}{2} V^2 \frac{\varepsilon_0 AK}{d} \left(\frac{1}{K+1} - 1 \right)$$
$$= \frac{1}{2} \frac{\varepsilon_0 AKV^2}{d} \frac{1-K-1}{K+1}$$
$$= \frac{1}{2} \frac{\varepsilon_0 AV^2}{d} \left(\frac{-K^2}{K+1} \right)$$

Question: The figure shows a circuit having eight resistances of 1 Ω each, labelled R_1 to R_8, and two ideal batteries with voltages $\varepsilon_1 = 12\ V$ and $\varepsilon_2 = 6\ V$



Which of the following statement(s) is(are) correct?

Options:

(a) The magnitude of current flowing through R_1 is 7.2 A.

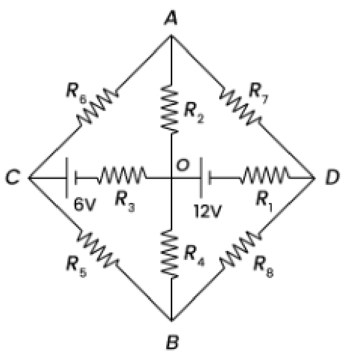
(b) The magnitude of current flowing through R_2 is 1.2 A.

(c) The magnitude of current flowing through R_3 is 4.8 A.

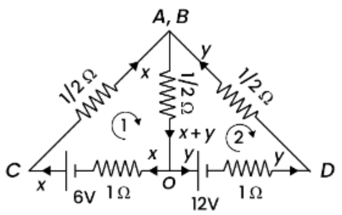
(d) The magnitude of current flowing through R_4 is 2.4 A.

Answer: (a, b, c, d)

Solution:



Point A and B are at same potential so they can be merged/folded.



For loop 1

$$-\frac{1}{2}x-\frac{1}{2}(x+y)-x+6=0$$

$$-2x-\frac{y}{2}=-6$$

$$-4x-y=-12$$

$$4x-y=-12$$

$$4x+y=12\ \ ...(1)$$

For loop 2

$$\frac{1}{2}y+1y+12+\frac{1}{2}(x+y)=0$$

$$\frac{3}{2}y+\frac{y}{2}+\frac{x}{2}=-12$$

$$2y+\frac{x}{2}=-12$$

$$4y+x=-24$$

$$4y+x-16x-4y$$

$$=-24-48$$

$$-15x=-72$$

$$x=\frac{72}{15}$$

$$4\left(\frac{72}{15}\right)+y=12$$

$$y=12-\frac{288}{15}$$

$$=\frac{180-288}{15}$$

$$=\frac{-108}{15}=-7.2A$$

Current in R_1 = 7.2 A

Current in R_2 $= \frac{x+y}{2} = \left(\frac{72}{15} - \frac{108}{15}\right)\frac{1}{2}$

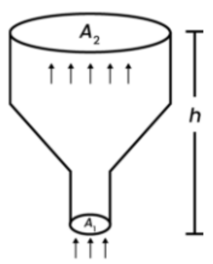
$= \frac{1}{2}\left(\frac{36}{15}\right) = \frac{2.4}{2}A$

$= 1.2A$

Current in $R_3 = x = 4.8A$

Current in R_5 $= \frac{1}{2}x = 2.4A$

Question: An ideal gas of density $\rho = 0.2\ kg\ m^{-3}$ enters a chimney of height h at the rate of $\alpha = 0.8\ kg\ s^{-1}$ from its lower end, and escapes through the upper end as shown in the figure. The
cross-sectional area of the lower end is $A_1 = 0.1\ m^2$ and the upper end is $A_2 = 0.4\ m^2$. The pressure and the temperature of the gas at the lower end are 600 Pa and 300 K, respectively, while its temperature at the upper end is 150 K. The chimney is heat insulated so that the gas undergoes adiabatic expansion. Take $g = 10\ ms^{-2}$ and the ratio of specific heats of the gas $\gamma = 2$. Ignore atmospheric pressure.



Which of the following statement(s) is(are) correct?

Options:

(a) The pressure of the gas at the upper end of the chimney is 300 Pa.

(b) The velocity of the gas at the lower end of the chimney is 40 ms^{-1} and at the upper end is 20 ms^{-1}.

(c) The height of the chimney is 590 m.

(d) The density of the gas at the upper end is 0.05 $kg\ m^{-3}$.

Answer: (b, c)

Solution:

$$\frac{\rho_1}{\rho_2} = \left(\frac{T_1}{T_2}\right)^{\frac{1}{\gamma-1}}$$

$$\Rightarrow \rho_2 = (0.2)\left(\frac{150}{300}\right)^{\frac{1}{1}} = 0.1kg/m^3$$

Rate of mass flow = 0.8 kg/s

$\Rightarrow$ Volume flow rate at top = 8 m^3s

$$\frac{P_2}{P_1}=\left(\frac{\rho_2}{\rho_1}\right)^{\gamma} \Rightarrow P_2=600\left(\frac{1}{2}\right)^{\gamma}$$
$$\Rightarrow P_2=150Pa$$

Velocity of gas at lower end $=\frac{V_1}{A_1}=\frac{0.8}{0.2\times0.1}=40\,m/s$

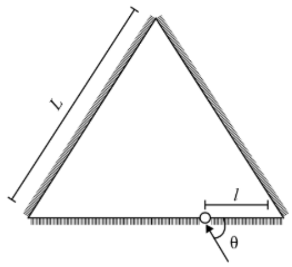
Velocity of gas at upper end $=\frac{0.8\times2}{0.2\times0.4}=20\,m/s$

By applying energy conservation

$$\frac{1}{2}\rho_1v_1^2+P_1=\frac{1}{2}\rho_2v_2^2+P_2+\rho_2gh_2$$

$$\Rightarrow\frac{1}{2}(0.2)(1600)+600=\frac{1}{2}(0.1)(400)+150+(0.1)(10)h_2$$
$$\Rightarrow h_2=590m$$

Question: Three plane mirrors form an equilateral triangle with each side of length L. There is a small hole at a distance $l > 0$ from one of the corners as shown in the figure. A ray of light is passed through the hole at an angle θ and can only come out through the same hole. The cross
section of the mirror configuration and the ray of light lie on the same plane.



Which of the following statement(s) is(are) correct?

Options:

(a) The ray of light will come out for $\theta = 30°$, for $0 < l < L$.

(b) There is an angle for $l=\frac{L}{2}$ at which the ray of light will come out after two reflections.

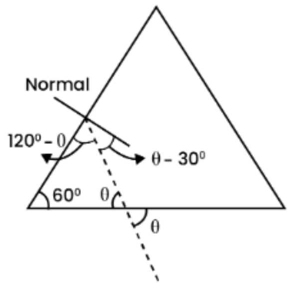
(c) The ray of light will NEVER come out for $\theta = 60°$, and $l = \frac{L}{3}$.

(d) The ray of light will come out for $\theta = 60°$, and $0 < l < \frac{L}{2}$ after six reflections

after six reflections.

Answer: (a, b)

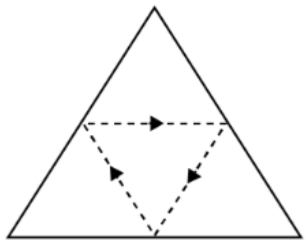
Solution:



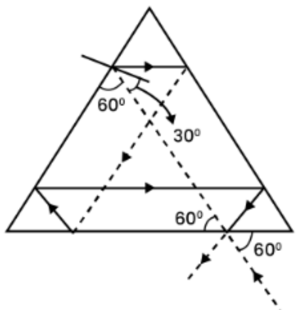
As we can see, for $\theta = 30°$, the ray will incident normally and hence will retrace its path.

$\Rightarrow$ (A) is correct

Considering the symmetry of the situation, we can have



$\Rightarrow$ (B) is correct



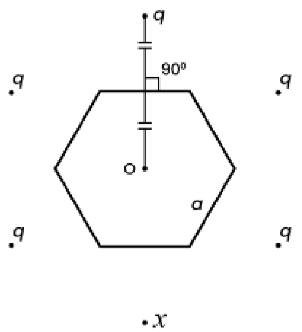
As is clear from the above diagram, ray comes out.

$\Rightarrow$ (C) is not correct

Also, as is clear from the above diagram, total number of reflections = 5.

$\Rightarrow$ (D) is not correct

Question: Six charges are placed around a regular hexagon of side length a as shown in the figure. Five of them have charge q, and the remaining one has charge x. The perpendicular from each charge to the nearest hexagon side passes through the center O of the hexagon and is bisected by the side.



Which of the following statement(s) is(are) correct in SI units?

Options:

(a) When $x = q$, the magnitude of the electric field at O is zero.

(b) When $x = -q$, the magnitude of the electric field at O is $\frac{q}{6\pi\varepsilon_0 a^2}$.

(c) When $x = 2q$, the potential at O is $\frac{7q}{4\sqrt{3}\pi\varepsilon_0 a}$.

(d) When $x = -3q$, the potential at O is $-\frac{3q}{4\sqrt{3}\pi\varepsilon_0 a}$

Answer: (a, b, c)

Solution:

When x = q, the situation is symmetric

$\Rightarrow$ Electric field at O would be zero.

$\Rightarrow$ (A) is correct.

When x = –q, we can think of x as q + (– 2q) $\Rightarrow$ Magnitude of electric field

$$\text{At } O = \frac{1}{4\pi\varepsilon_0}\frac{(2q)}{\left(2\times\frac{\sqrt{3}a}{2}\right)^2}$$

$$= \frac{1}{4\pi\varepsilon_0}\frac{2q}{3a^2} = \frac{q}{6\pi\varepsilon_0 a^2}$$

(B) is correct

For x = 2q potential at O is

$$V_0 = 6\times\frac{1}{4\pi\varepsilon_0}\times\frac{q}{\sqrt{3}a} + \frac{1}{4\pi\varepsilon_0}\frac{q}{\sqrt{3}a}$$

$$= \frac{7q}{4\sqrt{3}\pi\varepsilon_0 a}$$

(C) is correct

For $x=-3q, V_0 = 2\times\frac{1}{4\pi\varepsilon_0}\times\frac{q}{\sqrt{3}a}=\frac{q}{2\sqrt{3}\pi\varepsilon_0 a}$

D is not correct

Question: The binding energy of nucleons in a nucleus can be affected by the pairwise Coulomb repulsion. Assume that all nucleons are uniformly distributed inside the nucleus. Let the binding energy of a proton be E_b^p and the binding energy of a neutron be E_b^n in the nucleus.

Which of the following statement(s) is(are) correct?

Options:

(a) $E_b^p - E_b^n$ is proportional to $Z(Z-1)$ where Z is the atomic number of the nucleus.

(b) $E_b^p - E_b^n$ is proportional to $A^{-1/3}$ where A is the mass number of the nucleus.

(c) $E_b^p - E_b^n$ is positive.

(d) E_b^p increases if the nucleus undergoes a beta decay emitting a positron.

Answer: (a, b, d)

Solution:

Total binding energy (without considering repulsions),

$E_b = \left[Zm_p + (A-Z)m_n - m_x\right]c^2$

Where, ${}_Z^A X$ is the nuclei under consideration.

Now, considering repulsion :

Number of proton pairs $-{}^Z C_2$

$\Rightarrow$ This repulsion energy $\propto \frac{Z(Z-1)}{2}\times\frac{1}{4\pi\varepsilon_0}\frac{e^2}{R}$

Where R is the radius of the nucleus

$\Rightarrow E_b^p - E_b^n \propto Z(Z-1)$ $\therefore$ there will be no repulsion term for neutrons.

Also, since $R - R_0 A^{1/3}$

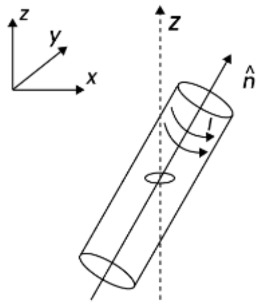
$E_b^p < E_b^n$

Because of repulsion among protons, $E_b^p < E_b^n$

Since in β^+ decay, number of protons decrease $\Rightarrow$ repulsion would decrease

$\Rightarrow E_b^p$ increases

Question: A small circular loop of area A and resistance R is fixed on a horizontal xy-plane with the center of the loop always on the axis $\hat{n}$ it of a long solenoid. The solenoid has m turns per unit length and carries current I counter clockwise as shown in the figure. The magnetic field due to the solenoid is in $\hat{n}$ direction. List-I gives time dependences of $\hat{n}$ in terms of a constant angular frequency ω. List-II gives the torques experienced by the circular loop at time $t = \frac{\pi}{6\omega}$. Let $\alpha = \frac{A^2\mu_0^2 m^2 I^2 \omega}{2R}$.



List I	List – II
(I) $\frac{1}{\sqrt{2}}\left(\sin\omega t\,\hat{j} + \cos\omega t\,\hat{k}\right)$	(P) 0
(II) $\frac{1}{\sqrt{2}}\left(\sin\omega t\,\hat{i} + \cos\omega t\,\hat{j}\right)$	(Q) $-\frac{\alpha}{4}\hat{i}$
(III) $\frac{1}{\sqrt{2}}\left(\sin\omega t\,\hat{i} + \cos\omega t\,\hat{k}\right)$	(R) $\frac{3a}{4}\hat{i}$
(IV) $\frac{1}{\sqrt{2}}\left(\cos\omega t\,\hat{j} + \sin\omega t\,\hat{k}\right)$	(S) $\frac{\alpha}{4}\hat{j}$
	(T) $-\frac{3\alpha}{4}\hat{i}$

Which one of the following options is correct?

Options:

(a) I $\rightarrow$ Q, II $\rightarrow$ P, III $\rightarrow$ S, IV $\rightarrow$ T

(b) I $\rightarrow$ S, II $\rightarrow$ T, III $\rightarrow$ Q, IV $\rightarrow$ P

(c) I $\rightarrow$ Q, II $\rightarrow$ P, III $\rightarrow$ S, IV $\rightarrow$ R

(d) I $\rightarrow$ T, II $\rightarrow$ Q, III $\rightarrow$ P, IV $\rightarrow$ R

Answer: (c)

Solution:

$$\boxed{\phi = BA\hat{k}.\hat{n}}$$

$$\phi = \frac{BA}{\sqrt{2}}\cos(\omega t)$$

$$\varepsilon = \frac{BA\omega}{\sqrt{2}}\sin(\omega t)$$

$$i=\frac{BA\omega}{\sqrt{2}R}\sin(\omega t)$$

$$\vec{m}=iA\hat{k}=\frac{BA^2\omega}{\sqrt{2}R}\sin(\omega t)\hat{k}$$

$$\vec{\tau}=\vec{m}\times\vec{B}=\frac{B^2A^2\omega}{\sqrt{2}R}\sin(\omega t)\left(\hat{k}\times\hat{n}\right)$$

$$=-\frac{B^2A^2\omega}{2R}\left[\hat{i}\right]\sin^2(\omega t)$$

$$\tau=-\frac{B^2A^2\omega}{2R}\left[\sin^2\left(\frac{\pi}{6}\right)\right]=\frac{-\alpha}{4}\hat{i}$$

$(1)\rightarrow Q$

$II.\phi=O$

$II\rightarrow P$

III. $\phi=\frac{BA}{\sqrt{2}}\cos(\omega t)$

$$i=\frac{BA\omega}{\sqrt{2}R}\sin(\omega t)$$

$$\vec{m}=\frac{BA^2\omega}{\sqrt{2}R}\sin(\omega t)\hat{k}$$

$$\vec{\tau}=\vec{m}\times\vec{B}=\frac{B^2A^2\omega}{\sqrt{2}\times\sqrt{2}R}\sin\omega t\left(\hat{k}\times\left(\sin\omega t\hat{i}+\cos\omega t\hat{k}\right)\right)$$

$$\vec{\tau}=\frac{B^2A^2\omega\sin(\omega t)}{2R}\sin(\omega t)\hat{j}$$

$$=\frac{B^2A^2\omega}{2R}\sin^2(\omega t)\hat{j}$$

$$=\frac{\alpha}{4}\hat{j}$$

III $\rightarrow$ **S**

IV. $\phi=\frac{BA}{\sqrt{2}}\sin(\omega t)$

$$i=-\frac{BA\omega}{\sqrt{2}R}\cos(\omega t)$$

$$\vec{m}=-\frac{BA^2\omega}{\sqrt{2}R}\cos\omega t\left(\hat{k}\right)$$

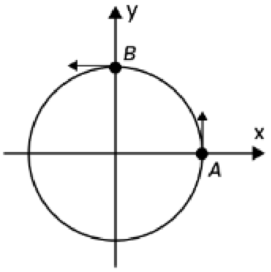
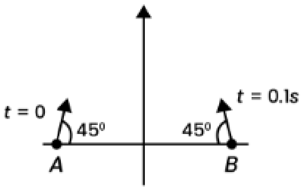
$$\vec{\tau}=\vec{m}\times\vec{B}=-\frac{B^2A^2\omega}{2R}\left(\hat{k}\times\hat{j}\right)\cos^2(\omega t)$$

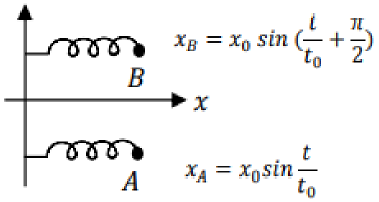
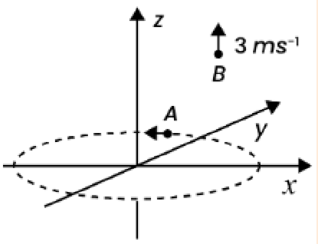
$$\tau=+\frac{B^2A^2\omega}{2R}\left(\hat{i}\right).\cos^2\left(\frac{\pi}{6}\right)$$

$$=+\frac{3}{4}\alpha\hat{i}$$

(IV) $\rightarrow$ R

Question: List I describes four systems, each with two particles A and B in relative motion as shown in figures. List II gives possible magnitudes of their relative velocities (in ms^{-1}) at time $t=\frac{\pi}{3}s$.

List-I	List-II
(I) A and B are moving on a horizontal circle of radius 1 m with uniform angular speed ω = 1 $rad\ s^{-1}$. The initial angular positions of A and B at time t = 0 are θ = 0 and θ = $\frac{\pi}{2}$, respectively. 	(P) $\frac{\sqrt{3}+1}{2}$
(II) Projectiles A and B are fired (in the same vertical plane) at $t=0$ and $t=0.1\ s$ respectively, with the same speed $v=\frac{5\pi}{\sqrt{2}}ms^{-1}$ and 45° from the horizontal plane. The initial separation between A and B is large enough so that they do not collide. (g = 10 ms^{-2}). 	(Q) $\frac{\left(\sqrt{3}-1\right)}{\sqrt{2}}$

(III) Two harmonic oscillators A and B moving in the x direction according to $x_A = x_0 \sin\frac{t}{t_0}$ and $x_B = x_0 \sin\left(\frac{t}{t_0} + \frac{\pi}{2}\right)$ respectively, starting from t = 0. Take x_0 = 1m, t_0 = 1 s.  	(R) $\sqrt{10}$
(IV) Particle A is rotating in a horizontal circular path of radius 1 m on the xy plane, with constant angular speed ω = 1 rad s^{-1}. Particle B is moving up at a constant speed 3 ms^{-1} in the vertical direction as shown in the figure. (Ignore gravity.)  	(S) $\sqrt{2}$
	(T) $\sqrt{25\pi^2 + 1}$

Which one of the following options is correct?

Options:

(a) I $\rightarrow$ R, II $\rightarrow$ T, III $\rightarrow$ P, IV $\rightarrow$ S

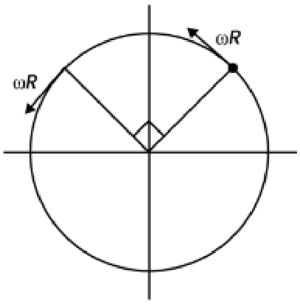
(b) I $\rightarrow$ S, II $\rightarrow$ P, III $\rightarrow$ Q, IV $\rightarrow$ R

(c) I $\rightarrow$ S, II $\rightarrow$ T, III $\rightarrow$ P, IV $\rightarrow$ R

(d) I $\rightarrow$ T, II $\rightarrow$ P, III $\rightarrow$ R, IV $\rightarrow$ S

Answer: (c)

Solution:



$v_{net} = \sqrt{2}\omega R = \sqrt{2}$

$1 \to S$

(II) $v_A\left(t = 0.13\right) = \frac{5\pi}{\sqrt{2}}\cos 45\hat{i}\text{ "}+\left[\frac{5\pi}{\sqrt{2}} \times \frac{1}{\sqrt{2}} - g \times 0.1\right]\hat{j}$

$= \frac{5\pi}{2}\hat{i} + \left(\frac{5\pi}{2} - 1\right)\hat{j}$

$v_B\left(t = 0.1\sec\right) = \frac{-5\pi}{2}\hat{i} + \left(\frac{5\pi}{2}\right)\hat{j}$

After t = 0.1, relative velocities should not change

$v_{rel}\left(t = 0.1\sec\right) = \left|5\pi\hat{i} - \hat{j}\right|$

$= \sqrt{25\pi^2 + 1}$

II $\to$ T

(III) $x = x_A - x_B$

$= x_0 \sin t - x_0 \sin\left(t + \frac{\pi}{2}\right)$

$= \sqrt{2}x_0 \sin\left(t - \frac{\pi}{4}\right)$

$v_{rel} = \frac{dx}{dt} = \sqrt{2}x_0 \cos\left(t - \frac{\pi}{4}\right)$

$= \sqrt{2}\cos\left(\frac{\pi}{3} - \frac{\pi}{4}\right)$

$= \sqrt{2} \times \frac{\sqrt{3}+1}{2\sqrt{2}} = \frac{\sqrt{3}+1}{2}$

III $\to$ P

(IV) $v_{rel} = \sqrt{3^2 + 1^2} = \sqrt{10}$

IV → R

Question: List I describes thermodynamic processes in four different systems. List II gives the magnitudes (either exactly or as a close approximation) of possible changes in the internal energy of the system due to the process.

List I	List II
(I) 10^{-3} kg of water at 100°C is converted to steam at the same temperature, at a pressure of 10^5 Pa. The volume of the system changes from 10^{-6} m^3 in the process. Latent heat of water = 2250 kJ/kg.	(P) 2 kJ
(II) 0.2 moles of a rigid diatomic ideal gas with volume V at temperature 500 K undergoes an isobaric expansion to volume 3 V. Assume R = 8.0 $J\ mol^{-1}\ K^{-1}$.	(Q) 7 kJ
(III) One mole of a monatomic ideal gas is compressed adiabatically from volume $V = \frac{1}{3} m^3$ and pressure 2 kPa to volume $\frac{V}{8}$.	(R) 4 kJ
(IV) Three moles of a diatomic ideal gas whose molecules can vibrate, is given 9 kJ of heat and undergoes isobaric expansion.	(S) 5 kJ
	(T) 3 kJ

Which of the following options is correct?

Options:

(a) I → T, II → R, III → S, IV → Q

(b) I → S, II → P, III → T, IV → P

(c) I → P, II → R, III → T, IV → Q

(d) I → Q, II → R, III → S, IV → T

Answer: (c)

Solution:

(i) $U = ML - P\Delta V$

$= 10^{-3} \times 2250 - 10^2 kP \times \left(10^{-3} - 10^{-6}\right) m^3$

$= 2.25 kJ - 0.1 kJ$

$= 2.15 kJ$

1 → P

(ii) $C_V = \frac{5R}{2}(\text{rigid diatomic})$

For isobaric expansion

$V \propto T$

$$\frac{V_1}{V_2} = \frac{T_1}{T_2} \Rightarrow \frac{V}{3V} = \frac{500}{T_2} \Rightarrow T_2 = 1500K$$

$$\Delta U = nC_V \Delta T = 0.2 \times \frac{5 \times 8}{2} \times (1500 - 500) J$$

$= 4kJ$

II $\rightarrow$ R

(III) Adiabatic expansion $\left(\gamma = \frac{5}{3}\right)$

$$P_1.V_1^\gamma = P_2 V_2^\gamma \Rightarrow (2kPa) \times V_0^{5/3} = P_2 \times \left(\frac{V_0}{8}\right)^{5/3}$$

$P_2 = 64kPa$

$\Delta U = nC_v \Delta T$

$$= \frac{3nR\Delta T}{2} = \frac{3}{2}(P_2V_2 - P_1V_1) = \frac{3}{2} \times \left(64 \times \frac{1}{3 \times 8} - 2 \times \frac{1}{3}\right)$$

$$= \frac{3}{2} \times \left(\frac{8}{3} - \frac{2}{3}\right) = 3kJ$$

III $\rightarrow$ T

(IV) For isobaric expansion.

$$\Delta U = nC_v \Delta T = \frac{7}{2} nR\Delta T$$

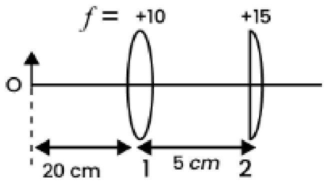
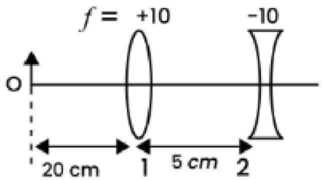
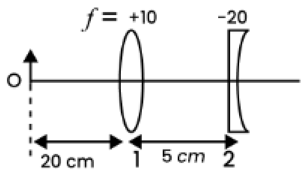
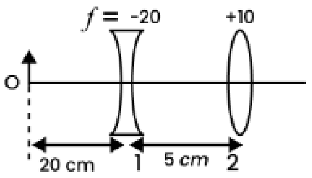
$$\Delta Q = nC_p \Delta T = \frac{9}{2} nR\Delta T$$

$$\frac{\Delta U}{\Delta Q} = \frac{7}{9}$$

$$\Delta U = \frac{7}{9} \Delta Q = 7kJ$$

IV $\rightarrow$ Q

Question: List I contains four combinations of two lenses (1 and 2) whose focal lengths (in cm) are indicated in the figures. In all cases, the object is placed 20 cm from the first lens on the left, and the distance between the two lenses is 5 cm. List II contains the positions of the final images.

List I	List II
(I) 	(P) Final image is formed at 7.5 cm on the right side of lens 2.
(II) 	(Q) Final image is formed at 60.0 cm on the right side of lens 2.
(III) 	(R) Final image is formed at 30.0 cm on the left side of lens 2.
(IV) 	(S) Final image is formed at 6.0 cm on the right side of lens 2.
	(S) Final image is formed at 30.0 cm on the right side of lens 2.

Which one of the following options is correct?

Options:

(a) I → P, II → R, III → Q, IV → T

(b) I → Q, II → P, III → T, IV → S

(c) I → P, II → T, III → R, IV → Q

(d) I → Q, II → S, III → Q, IV → R

Answer: (a)

Solution:

u = -20 cm

f = + 10 cm

$$\frac{1}{v}+\frac{1}{20}=\frac{1}{10}$$

$$\frac{1}{v}=\frac{1}{10}-\frac{1}{20}$$
$$\frac{1}{v}=\frac{1}{20}$$
$$v=20\,cm$$

v = 20 cm

u = +15 cm

f = + 15 cm

$$\frac{1}{v}-\frac{1}{u}=\frac{1}{f}$$
$$\Rightarrow \frac{1}{v}-\frac{1}{15}=\frac{1}{15}$$
$$\frac{1}{v}=\frac{2}{15}$$

v = 7.5 cm (from lens 2)

I $\rightarrow$ P

(II) u = -20 cm, f + 10 cm

$$\frac{1}{v}+\frac{1}{20}=\frac{1}{10}$$
$$\frac{1}{v}=\frac{1}{10}-\frac{1}{20}\Rightarrow v=20\,cm$$
$$u=+15\,cm$$

f = -10 cm

$$\frac{1}{v}-\frac{1}{u}=\frac{1}{f}\Rightarrow \frac{1}{v}-\frac{1}{15}=\frac{-1}{10}$$
$$\frac{1}{v}=\frac{-1}{10}+\frac{1}{15}=\frac{-3+2}{30}$$
$$\frac{1}{v}=-\frac{1}{30}$$

v = -30 cm

II $\rightarrow$ R

(III) u = -20

f = + 10 cm

$$\frac{1}{v}+\frac{1}{20}=\frac{1}{10}$$
$$\frac{1}{v}=\frac{1}{10}-\frac{1}{20}=\frac{1}{20}$$
$$v=20\,cm$$

$$\Rightarrow u=15cm$$
$$f=-20\,cm$$
$$\frac{1}{v}-\frac{1}{15}=\frac{-1}{20}$$

$$\frac{1}{v}=-\frac{1}{20}+\frac{1}{15}=\frac{-3+4}{60}=\frac{1}{60}$$
$$v=60\,cm$$

III→Q

(IV) u = -20 cm

f = -20

$$\Rightarrow\frac{1}{v}-\frac{1}{u}=\frac{1}{f}$$
$$\Rightarrow\frac{1}{v}+\frac{1}{20}=\frac{-1}{20}$$

v = -10 cm

u = -15 cm

f = 10 cm

$$\frac{1}{v}-\frac{1}{u}=\frac{1}{f}$$
$$\Rightarrow\frac{1}{v}+\frac{1}{15}=\frac{1}{10}$$

$$\Rightarrow\frac{1}{v}=\frac{1}{10}-\frac{1}{15}$$
$$=\frac{3-2}{30}=\frac{1}{30}$$
$$v=30\,cm$$

IV → T

JEE-Advanced-28-08-2022
[Paper-1]

CHEMISTRY

Question: 2 mol of Hg(g) is combusted in a fixed volume bomb calorimeter with excess of O_2 at 298 K and 1 atm into HgO(s). During the reaction, temperature increases from 298.0 K to 312.8 K. If heat capacity of the bomb calorimeter and enthalpy of formation of Hg(g) are 20.00 kJ K^{-1} and 61.32 kJ mol^{-1} at 298 K, respectively, the calculated standard molar enthalpy of formation of HgO(s) at 298 K is X kJ mol^{-1}. The value of |X| is _______.
[Given: Gas constant R = 8.3 J K^{-1} mol^{-1}]

Answer: 90.39

Solution:

$\Delta T = 312.8 - 298 = 14.8\ K$

$\Rightarrow$ Heat released at constant volume = $C\Delta T = 20 \times 14.8 = 296\ kJ$

$2Hg(g) + O_2(g) \rightarrow 2HgO(s)$

$\Delta n_g = -3$

$\Rightarrow$ heat released at constant pressure (ΔH)

$\Delta H = \Delta U + \Delta n_g RT$

$\Delta H = -296 + (-3) \times 8.3 \times 298 \times 10^{-3}$

$\Delta H = -303.42\ kJ$

$\Rightarrow -303.42 = 2\Delta H_f^o(HgO) - 2\Delta H_f^o(Hg)$

$$\Rightarrow \Delta H_f^o(HgO) = \frac{-303.42 + 2(61.32)}{2}$$

$= -90.39\ kJ$

So, |X| = 90.39

Question: The reduction potential (E^0, in V) of MnO_4^-(aq)/Mn(s) is ______.
[Given: $E^o_{(MnO_4^-(aq)/MnO_2(s))} = 1.68\ V; E^o_{(MnO_2(s)/Mn^{2+}(aq))} = 1.21\ V; E^o_{(Mn^{2+}(aq)/Mn(s))} = -1.03\ V$]

Answer: 0.77

Solution:

$\overset{+7}{M}nO_4^- + 3e^- \rightarrow \overset{+4}{M}nO_2 \qquad E_1^o = 1.68\ V$

$\overset{+4}{M}nO_2 + 2e^- \rightarrow \overset{+2}{M}n^{2+} \qquad E_2^o = 1.21\ V$

$\overset{+2}{M}n^{2+} + 2e^- \rightarrow \overset{0}{M}n \qquad E_3^o = -1.03\ V$

$\overset{+7}{M}nO_4^- + 7e^- \rightarrow \overset{0}{M}n \qquad E_4^o = ?$

$$\Delta G_4^o = \Delta G_1^o + \Delta G_2^o + \Delta G_3^o$$
$$\Rightarrow -7FE_4^o = -3FE_1^o - 2FE_2^o - 2FE_3^o$$
$$\Rightarrow E_4^o = \frac{3\times1.68 + 2\times1.21 - 2\times1.03}{7} = 0.77\ V$$

Question: A solution is prepared by mixing 0.01 mol each of H_2CO_3, $NaHCO_3$, Na_2CO_3, and NaOH in 100 mL of water. *p*H of the resulting solution is _______.
[Given: pK_{a_1} and pK_{a_2} of H_2CO_3 are 6.37 and 10.32, respectively; log 2 = 0.30]

Answer: 10.02

Solution:

$$\underset{0.01}{NaOH} + \underset{0.01}{H_2CO_3} \rightarrow \underset{0.01}{NaHCO_3} + H_2O$$

So, in the resultant mixture $\left.\begin{matrix} NaHCO_3 \rightarrow 0.02 \\ Na_2CO_3 \rightarrow 0.01 \end{matrix}\right\}$ Buffer

$$\text{pH} = pK_{a_2} + \log\frac{0.01}{0.02}$$
$$= 10.32 - \log 2 = 10.02$$

Question: The treatment of an aqueous solution of 3.74 g of $Cu(NO_3)_2$ with excess KI results in a brown solution along with the formation of a precipitate. Passing H_2S through this brown solution gives another precipitate **X**. The amount of **X** (in g) is ________.
[Given: Atomic mass of H = 1, N = 14, O = 16, S = 32, K = 39, Cu = 63, I = 127]

Answer: 0.32

Solution:

$$\underset{n_f=1}{Cu^{2+}} + I^- \rightarrow Cu_2I_2 + \underset{n_f=2}{I_2}$$
$$\underset{n_f=2}{I_2} + H_2S \rightarrow \underset{n_f=2}{S} + HI$$

as n_f of I_2 is same in both reactions so equivalent of $Cu(NO_3)_2$ = equivalent of S

$$\Rightarrow \frac{3.74}{187}\times1 = \frac{X}{32}\times2$$
$$\Rightarrow X = 0.32g$$

Question: Dissolving 1.24 g of white phosphorous in boiling NaOH solution in an inert atmosphere gives a gas **Q**. The amount of $CuSO_4$ (in g) required to completely consume the gas **Q** is _______.
[Given: Atomic mass of H = 1, O = 16, Na = 23, P = 31, S = 32, Cu = 63]

Answer: 2.38

Solution:

$$3H_2O + 3NaOH + \underset{1.24g}{P_4} \rightarrow PH_3 + 3NaH_2PO_2$$

$\Rightarrow$ moles of $P_4 = \frac{1.24}{124} = 0.01$

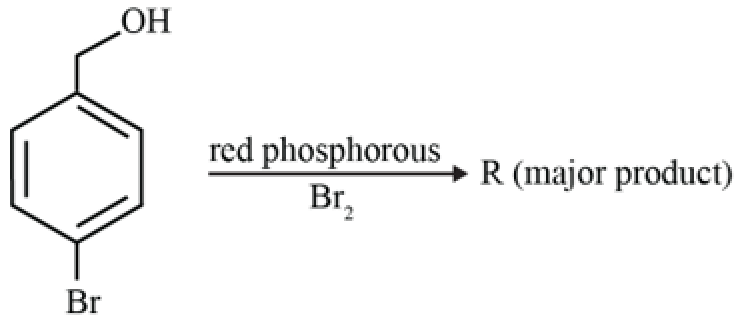
$$\underset{0.01}{2PH_3} + 3CuSO_4 \rightarrow Cu_3P_2 + 3H_2SO_4$$

$\Rightarrow$ moles of $CuSO_4$ needed $= \frac{3}{2} \times 0.01 = 0.015$

Weight $= 0.015 \times 159 = 2.385\ g$

= 2.38 g

Question: Consider the following reaction.



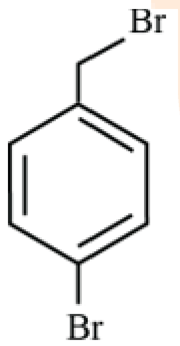
On estimation of bromine in 1.00 g of **R** using Carius method, the amount of AgBr formed (in g) is ________.

[Given: Atomic mass of H = 1, C = 12, O = 16, P = 31, Br = 80, Ag = 108]

Answer: 1.50

Solution:

R $\rightarrow$

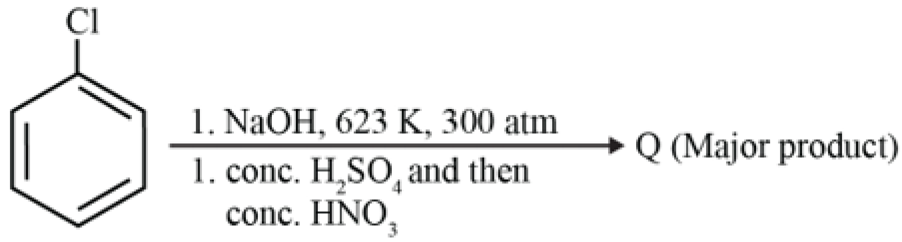


1 g

Moles of $AgBr = 2\times$ moles of $R = 2\times\frac{1}{250} = 0.008$

Mass of $AgBr = 0.008 \times 188 = 1.504 = 1.50\ g$

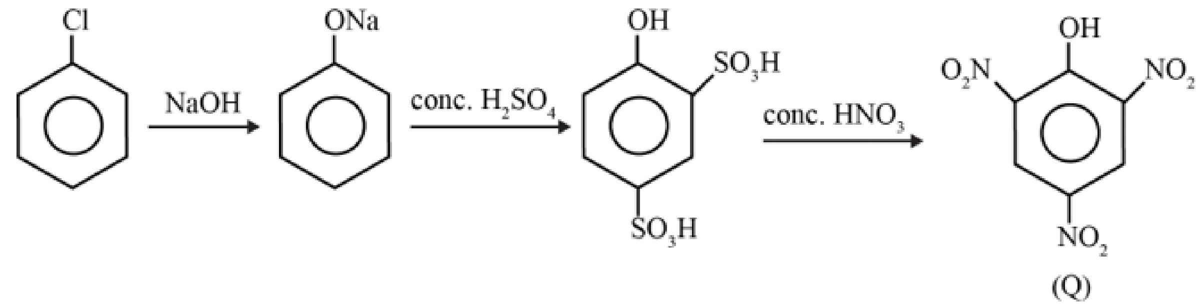
Question: The weight percentage of hydrogen in **Q**, formed in the following reaction sequence, is ________.



[Given: Atomic mass of H = 1, C = 12, N = 14, O = 16, S = 32, Cl = 35]

Answer: 1.31

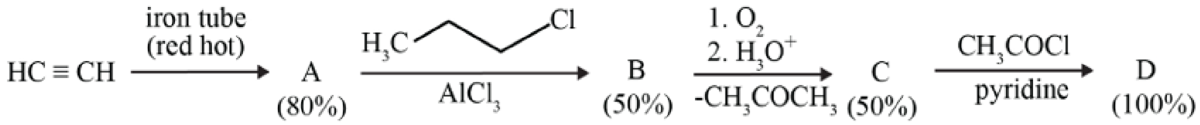
Solution:



$$\%H = \frac{\text{weight of 'H'}}{\text{molecular weight}} \times 100$$

$$\%H = \frac{3}{229} \times 100 = 1.31$$

Question: If the reaction sequence given below is carried out with 15 moles of acetylene, the amount of the product **D** formed (in g) is ________.

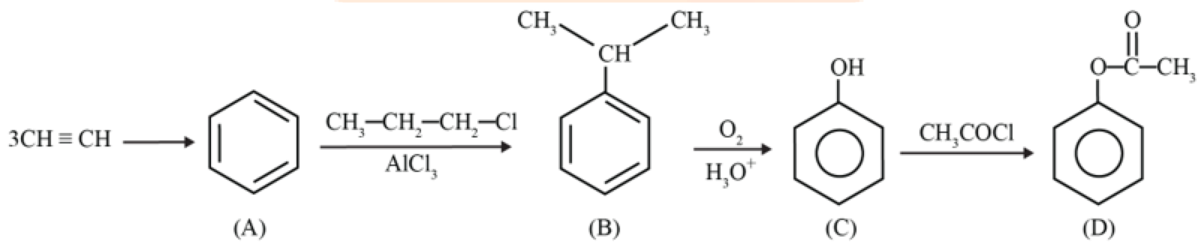


The yields of **A**, **B**, **C** and **D** are given in parentheses.

[Given: Atomic mass of H = 1, C = 12, O = 16, Cl = 35]

Answer: 136.00

Solution:



Actual moles of 'D' $= 5 \times \left(\frac{80}{100} \times \frac{50}{100} \times \frac{50}{100} \times \frac{100}{100} \right) = 1$ mol

Weight = 136 g

Question: For diatomic molecules, the correct statement(s) about the molecular orbitals formed by the overlap of two $2p_z$ orbitals is (are)

Options:

(a) σ orbital has a total of two nodal planes.

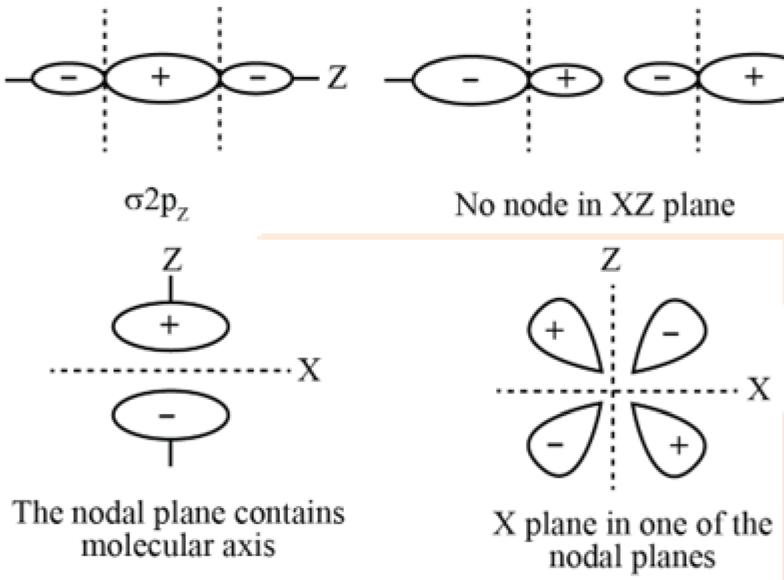
(b) σ^* orbital has one node in the xz-plane containing the molecular axis.

(c) π orbital has one node in the plane which is perpendicular to the molecular axis and goes through the center of the molecule.

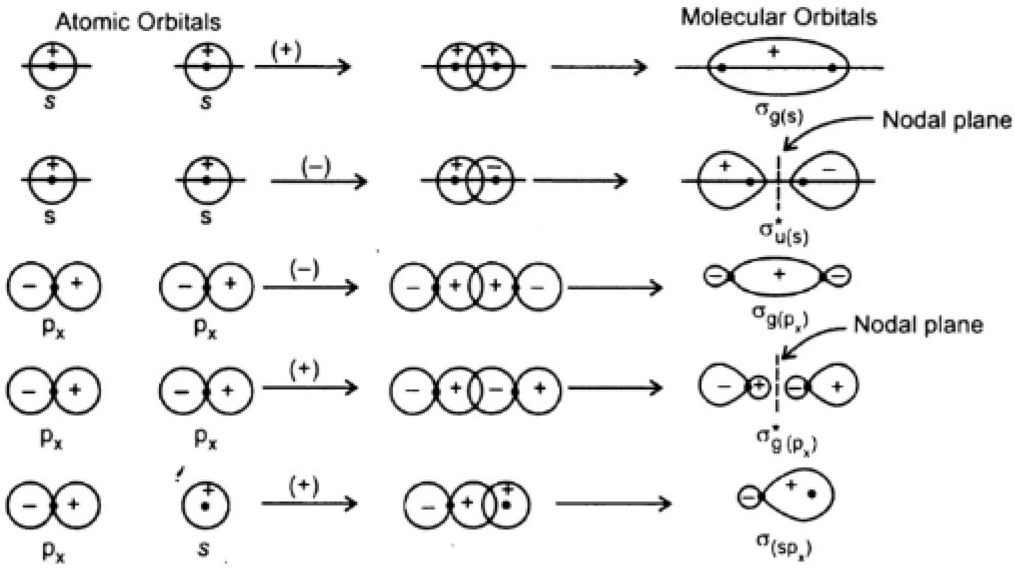
(d) π^* orbital has one node in the xy-plane containing the molecular axis.

Answer: (a), (d) or (d)

Solution:



Note- Answer can be only 'D' also



Question: The correct option(s) related to adsorption processes is(are)

Options:

(a) Chemisorption results in a unimolecular layer.

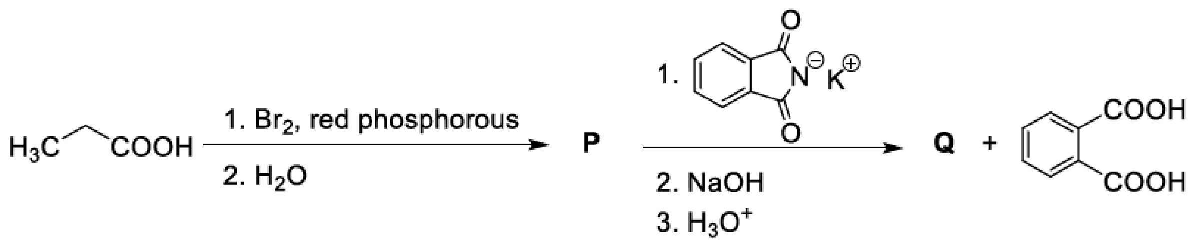
(b) The enthalpy change during physisorption is in the range of 100 to 140 kJ mol^{-1}.

(c) Chemisorption is an endothermic process
(d) Lowering the temperature favors physisorption processes.
Answer: (a), (d)
Solution:
1. Chemisorption is unimolecular layer, physisorption is multimolecular layer
2. Both chemisorption and physisorption are exothermic and ΔH for physisorption is 20-40 kJ mol^{-1} and 80 – 240 kJ mol^{-1}.
3. Physisorption is favored at low temperature.

Question: The electrochemical extraction of aluminum from bauxite ore involves
Options:
(a) the reaction of Al_2O_3 with coke (C) at a temperature > 2500 ºC.
(b) the neutralization of aluminate solution by passing CO_2 gas to precipitate hydrated alumina ($Al_2O_3 \cdot 3H_2O$).
(c) the dissolution of Al_2O_3 in hot aqueous NaOH.
(d) the electrolysis of Al_2O_3 mixed with Na_3AlF_6 to give Al and CO_2.
Answer: (b), (c), (d)
Solution: In electrochemical reduction such high temperature is not needed
$Al_2O_3 + NaOH \rightarrow NaAlO_2$
$NaAlO_2 + CO_2 \rightarrow Al_2O_3.3H_2O$

Question: The treatment of galena with HNO_3 produces a gas that is
Options:
(a) paramagnetic
(b) bent in geometry
(c) an acidic oxide
(d) colorless
Answer: (a), (d)
Solution:
$3PbS \downarrow + 8HNO_3 \rightarrow 3Pb^{2+} + 6NO_3^- + 3S \downarrow + 2NO \uparrow + 4H_2O$
NO is paramagnetic and colourless gas

Question: Considering the reaction sequence given below, the correct statement(s) is(are)

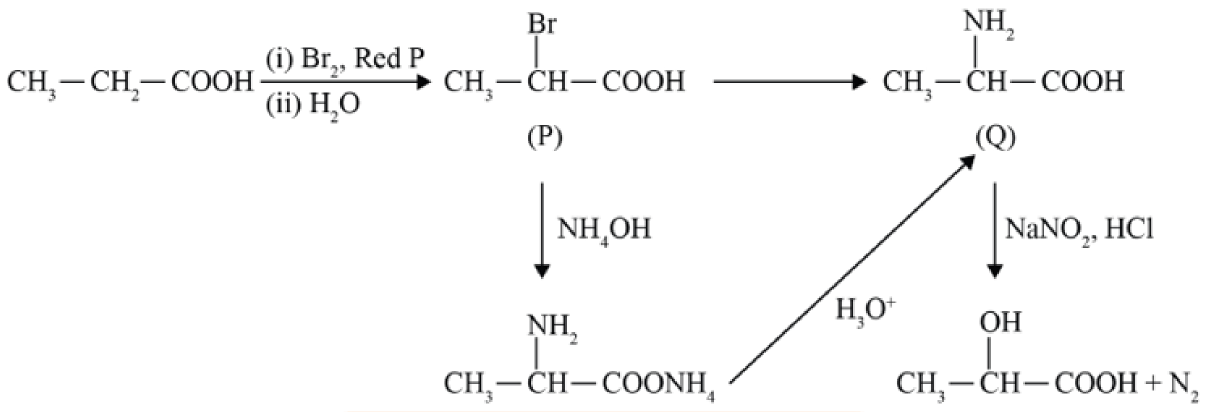


Options:
(a) **P** can be reduced to a primary alcohol using $NaBH_4$.
(b) Treating **P** with conc. NH_4OH solution followed by acidification gives **Q**.
(c) Treating **Q** with a solution of $NaNO_2$ in aq. HCl liberates N_2.

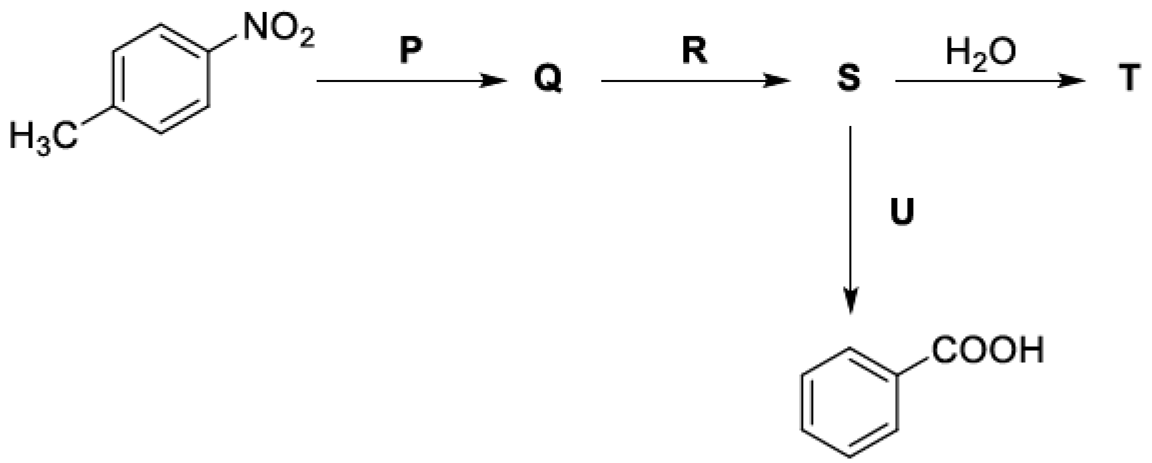
(d) **P** is more acidic than CH_3CH_2COOH.

Answer: (b), (c), (d)

Solution:



Question: Considering the following reaction sequence,



the correct option(s) is(are)

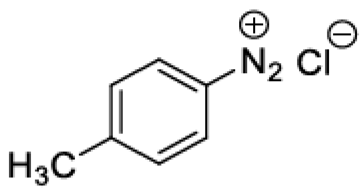
Options:

(a)

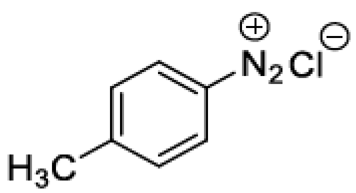
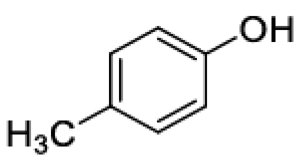
P = H_2/Pd, ethanol R = $NaNO_2$/HCl U = 1. H_3PO_2
2. $KMnO_4$ – KOH, heat

(b)

P = Sn/HCl R = HNO_2

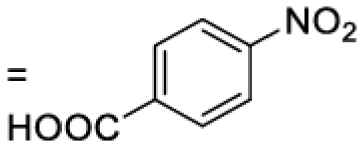
S = 

(c)

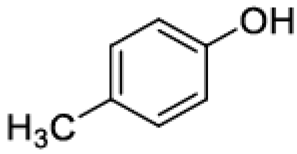
S =  **T =** 

U = 1. CH_3CH_2OH
2. $KMnO_4$ – KOH,
heat

(d)

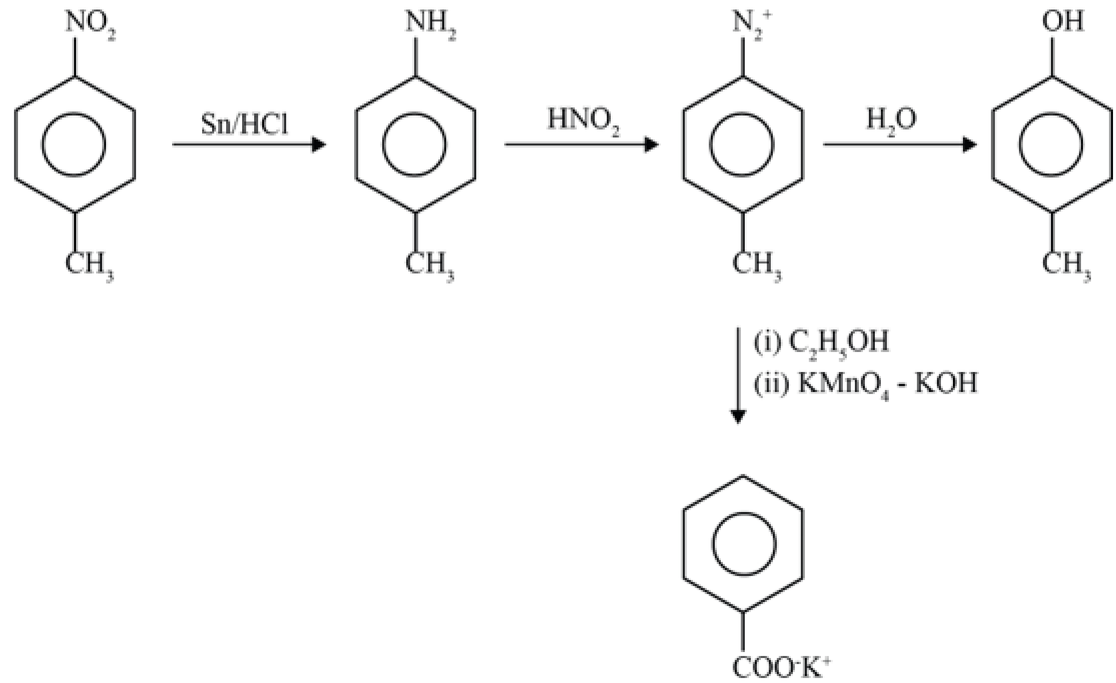
Q = 

R = H_2/Pd, ethanol

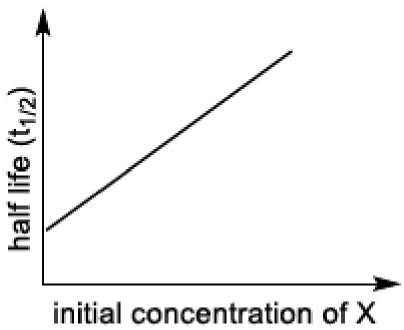
T = 

Answer: (b) or (a), (b), (c)

Solution:



Question: Match the rate expressions in LIST-I for the decomposition of X with the corresponding profiles provided in LIST-II. X_s and k are constants having appropriate units.

LIST-I	LIST-II
(I) rate = $\frac{k[X]}{X_s+[X]}$ under all possible initial concentrations of X	(P) 
(II) rate = $\frac{k[X]}{X_s+[X]}$ where initial concentrations of X are much less than X_s	(Q)

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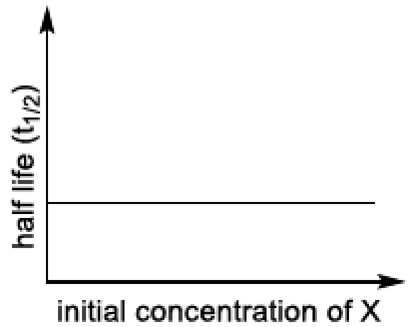
<tr><td></td><td>
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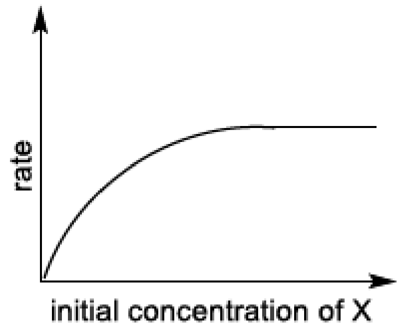
<tr><td>(III) rate $=\dfrac{k[X]}{X_s+[X]}$
where initial concentrations of X are much higher than X_s</td><td>(R)

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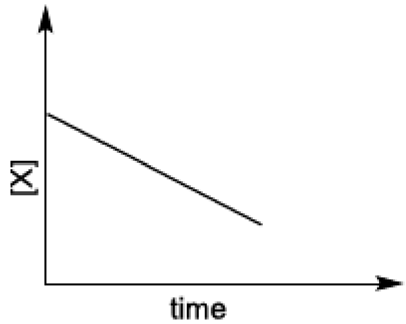
<tr><td>(IV) rate $=\dfrac{k[X]^2}{X_s+[X]}$
where initial concentration of X is much higher than X_s</td><td>(S)

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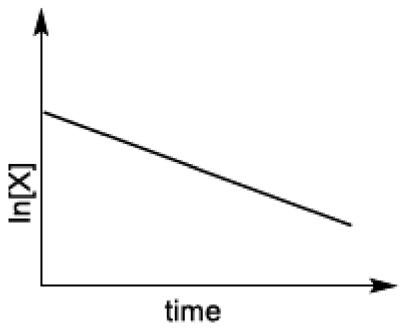
<tr><td></td><td>(T)

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Options:

(a) I→ P; II→ Q; III → S; IV → T

(b) I→ R; II→ S; III → S; IV → T

(c) I→ P; II→ Q; III → Q; IV → R

(d) I→ R; II→ S; III → Q; IV → R

Answer: (a)

Solution:

$$II \rightarrow r = \frac{K[X]}{X_s + [X]} \approx \frac{K[X]}{X_s} \quad \text{So first order}$$

$$III \rightarrow r = \frac{K[X]}{X_s + [X]} \approx K \quad \text{So zero order}$$

$$IV \rightarrow r = \frac{K[X]^2}{X_s + [X]} \approx K[X] \quad \text{So first order}$$

Question: LIST-I contains compounds and LIST-II contains reactions

LSIT-I	**LIST-II**
(I) H_2O_2	(P) $Mg(HCO_3)_2 + Ca(OH)_2 \rightarrow$
(II) $Mg(OH)_2$	(Q) $BaO_2 + H_2SO_4 \rightarrow$
(III) $BaCl_2$	(R) $Ca(OH)_2 + MgCl_2 \rightarrow$
(IV) $CaCO_3$	(S) $BaO_2 + HCl \rightarrow$
	(T) $Ca(HCO_3)_2 + Ca(OH)_2 \rightarrow$

Match each compound in LIST-I with its formation reaction(s) in LIST-II, and choose the correct option

Options:

(a) I → Q; II → P; III → S; IV → R

(b) I → T; II → P; III → Q; IV → R

(c) I → T; II → R; III → Q; IV → P

(d) I → Q; II → R; III → S; IV → P

Answer: (d)

Solution:

I → Q, S

II → P, R

III → S

IV → P, T

Question: LIST-I contains metal species and LIST-II contains their properties.

LIST-I	**LIST-II**
(I) $[Cr(CN)_6]^{4-}$	(P) t_{2g} orbitals contain 4 electrons
(II) $[RuCl_6]^{2-}$	(Q) μ(spin-only) = 4.9 BM
(III) $[Cr(H_2O)_6]^{2+}$	(R) low spin complex ion
(IV) $[Fe(H_2O)_6]^{2+}$	(S) metal ion in 4+ oxidation state
	(T) d^4 species

[Given: Atomic number of Cr = 24, Ru = 44, Fe = 26]

Match each metal species in LIST-I with their properties in LIST-II, and choose the correct option

Options:

(a) I → R, T; II → P, S; III → Q, T; IV → P, Q

(b) I → R, S; II → P, T; III → P, Q; IV → Q, T

(c) I → P, R; II → R, S; III → R, T; IV → P, T
(d) I → Q, T; II → S, T; III → P, T; IV → Q, R
Answer: (a)
Solution:

(I) Cr^{2+} CN = 6 SFL [ ][ ]
d^4

$\mu = \sqrt{8}\ B.M.$ [↑↓][↑][↑] t_{2g}
P, R, T so, $\cancel{B}$

(II) $[RuCl_6]^{2-}$ [ ][ ]

$Ru^{+4} = 4d^4 5s^0$; low spin complex [↑↓][↑][↑] t_{2g}
P, R, S, T

(III) $Cr^{2+} = d^4$ [↑][ ]

[↑][↑][↑]

$\mu = \sqrt{24}\ B.M.$
= 4.9
Q, T so, $\cancel{C}$ $\cancel{B}$

(IV) $Fe^{2+} = d^6$ [↑][↑]

[↑↓][↑][↑]

$\mu = \sqrt{24}\ B.M.$
= 4.9
P, Q, so, (a) is correct option

Question: Match the compounds in LIST-I with the observations in LIST-II, and choose the correct option.

LIST-I	LIST-II
(I) Aniline	(P) Sodium fusion extract of the compound on boiling with $FeSO_4$, followed by acidification with conc. H_2SO_4, gives Prussian blue color.
(II) *o*-Cresol	(Q) Sodium fusion extract of the compound on treatment with sodium nitroprusside gives blood red color.
(III) Cysteine	(R) Addition of the compound to a saturated solution of $NaHCO_3$ results in effervescence.

(IV) Caprolactam	(S) The compound reacts with bromine water to give a white precipitate.
	(T) Treating the compound with neutral $FeCl_3$ solution produces violet color.

Options:

(a) I→P,Q; II→S; III→Q,R; IV→P
(b) I→P; II→R,S; III→R; IV→Q,S
(c) I→Q,S; II→P,T; III→P; IV→S
(d) I→P,S; II→T; III→Q,R; IV→P

Answer: (d)

Solution:

I – P, S
II – T
III – Q, R
IV – P

Aniline converts to CN^- in Lassaigne test
Cysteine converts to SCN^- & S^{2-} in Lassaigne test
Aniline gives white ppt of 2, 4, 6 tribromo aniline with Bromine water
o-cresol gives violet colouration with neutral $FeCl_3$

JEE-Advanced-28-08-2022
[Paper-1]

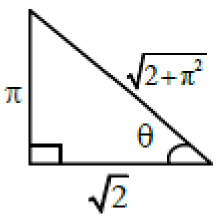
MATHEMATICS

Question: Considering only the principal values of the inverse trigonometric functions, the value of $\frac{3}{2}\cos^{-1}\sqrt{\frac{2}{2+\pi^2}}+\frac{1}{4}\sin^{-1}\frac{2\sqrt{2}\pi}{2+\pi^2}+\tan^{-1}\frac{\sqrt{2}}{\pi}$ is _____.

Answer: 2.35 or 2.36

Solution:

$$\cos^{-1}\sqrt{\frac{2}{2+\pi^2}}=\tan^{-1}\frac{\pi}{\sqrt{2}}$$



$$\sin^{-1}\left(\frac{2\sqrt{2}\pi}{2+\pi^2}\right)=\sin^{-1}\left(\frac{2\times\frac{\pi}{\sqrt{2}}}{1+\left(\frac{\pi}{\sqrt{2}}\right)^2}\right)$$

$$=\pi-2\tan^{-1}\left(\frac{\pi}{\sqrt{2}}\right)$$

(As, $\sin^{-1}\left(\frac{2x}{1+x^2}\right)=\pi-2\tan^{-1}x, x\geq 1$)

and $\tan^{-1}\frac{\sqrt{2}}{\pi}=\cot^{-1}\left(\frac{\pi}{\sqrt{2}}\right)$

$$\therefore \text{ Expression }=\frac{3}{2}\left(\tan^{-1}\frac{\pi}{\sqrt{2}}\right)+\frac{1}{4}\left(\pi-2\tan^{-1}\frac{\pi}{\sqrt{2}}\right)+\cot^{-1}\left(\frac{\pi}{\sqrt{2}}\right)$$

$$=\left(\frac{3}{2}-\frac{2}{4}\right)\tan^{-1}\frac{\pi}{\sqrt{2}}+\frac{\pi}{4}+\cot^{-1}\frac{\pi}{\sqrt{2}}$$

$$=\left(\tan^{-1}\frac{\pi}{\sqrt{2}}+\cot^{-1}\frac{\pi}{\sqrt{2}}\right)+\frac{\pi}{4}$$

$$=\frac{\pi}{2}+\frac{\pi}{4}=\frac{3\pi}{4}$$

$=2.35$ or 2.36

Question: Let α be a positive real number. Let $f:\mathbb{R}\to\mathbb{R}$ and $g:(\alpha,\infty)\to\mathbb{R}$ be the functions defined by $f(x)=\sin\left(\frac{\pi x}{12}\right)$ and $g(x)=\frac{2\log_e\left(\sqrt{x}-\sqrt{\alpha}\right)}{\log_e\left(e^{\sqrt{x}}-e^{\sqrt{\alpha}}\right)}$. Then the value of $\lim\limits_{x\to\alpha^+} f(g(x))$ is ___.

Answer: 0.50

Solution:

$$\lim_{x\to\alpha^+}\frac{2\ln\left(\sqrt{x}-\sqrt{\alpha}\right)}{\ln\left(e^{\sqrt{x}}-e^{\sqrt{\alpha}}\right)}\quad\left(\frac{0}{0}\text{ form}\right)$$

$\therefore$ Using L hospital rule,

$$=2\lim_{x\to\alpha^+}\frac{\left(\frac{1}{\sqrt{x}-\sqrt{\alpha}}\right)\cdot\frac{1}{2\sqrt{x}}}{\left(\frac{1}{e^{\sqrt{x}}-e^{\sqrt{\alpha}}}\right)\cdot e^{\sqrt{x}}\cdot\frac{1}{2\sqrt{x}}}$$

$$=\frac{2}{e^{\sqrt{\alpha}}}\lim_{x\to\alpha^+}\frac{\left(e^{\sqrt{x}}-e^{\sqrt{\alpha}}\right)}{\sqrt{x}-\sqrt{\alpha}}\quad\left(\frac{0}{0}\right)$$

$$=\frac{2}{e^{\sqrt{\alpha}}}\lim_{x\to\alpha^+}\frac{\left(e^{\sqrt{x}}\cdot\frac{1}{2\sqrt{x}}-0\right)}{\left(\frac{1}{2\sqrt{x}}-0\right)}=2$$

So, $\lim\limits_{x\to\alpha^+} f(g(x))=\lim\limits_{x\to\alpha^+} f(2)$

$=f(2)=\sin\frac{\pi}{6}=\frac{1}{2}$

$=0.50$

Question: In a study about a pandemic, data of 900 persons was collected. It was found that
190 persons had symptom of fever,
220 persons had symptom of cough,
220 persons had symptom of breathing problem,
330 persons had symptom of fever or cough or both,
350 persons had symptom of cough or breathing problem or both,
340 persons had symptom of fever or breathing problem or both,
30 persons had all three symptoms (fever, cough and breathing problem).
If a person is chosen randomly from these 900 persons, then the probability that the person has at most one symptom is ______________.

Answer: 0.80

Solution:

$n(U)=900$

Let $A\equiv$ fever, $B\equiv$ Cough

$C\equiv$ Breathing problem

$\therefore n(A)=190,\ n(B)=220,\ n(C)=220$

$n(A \cup B) = 330,\ n(B \cup C) = 350,$

$n(A \cup C) = 340,\ n(A \cap B \cap C) = 30$

Now, $n(A \cup B) = n(A) + n(B) - n(A \cap B)$

$\Rightarrow 330 = 190 + 220 - n(A \cap B)$

$\Rightarrow n(A \cap B) = 80$

Similarly,

$350 = 220 + 220 - n(B \cap C)$

$\Rightarrow n(B \cap C) = 90$

and $340 = 190 + 220 - n(A \cap C)$

$\Rightarrow n(A \cap C) = 70$

$\therefore n(A \cup B \cup C) = (190 + 220 + 220) - (80 + 90 + 70) + 30$

$= 660 - 240 = 420$

$\Rightarrow$ Number of person without any symptom

$= n(\cup) - n(A \cup B \cup C)$

$= 900 - 420 = 480$

Now, number of person suffering from exactly one symptom

$= (n(A) + n(B) + n(C)) - 2(n(A \cap B) + n(B \cap C) n(C \cap A)) + 3n(A \cap B \cap C)$

$= (190 + 220 + 220) - 2(80 + 90 + 7) + 3(30)$

$= 630 - 480 + 90 = 240$

$\therefore$ Number of person suffering from atmost one symotom

$= 480 + 240 = 720$

$\Rightarrow$ Probability $= \dfrac{720}{900} = \dfrac{8}{10} = \dfrac{4}{5} = 0.80$

Question: Let z be a complex number with non-zero imaginary part. If $\dfrac{2 + 3z + 4z^2}{2 - 3z + 4z^2}$ is a real number, then the value of $|z|^2$ is ____

Answer: 0.50

Solution:

Given that,

$z \neq \overline{z}$

Let $\alpha = \dfrac{2 + 3z + 4z^2}{2 - 3z + 4z^2} = \dfrac{(2 - 3z + 4z^2) + 6z}{2 - 3z + 4z^2}$

$\therefore \alpha = 1 + \dfrac{6z}{2 - 3z + 4z^2}$

If α is real number, then

$\alpha = \overline{\alpha}$

$\Rightarrow \dfrac{z}{2 - 3z + 4z^2} = \dfrac{\overline{z}}{2 - 3\overline{z} + 4\overline{z}^2}$

$\therefore 2\left(z-\bar{z}\right)=4z\bar{z}\left(z-\bar{z}\right)$

$\Rightarrow \left(z-\bar{z}\right)\left(2-4z\bar{z}\right)=0$

As $z \neq \bar{z}$ (Given)

$\Rightarrow z\bar{z}=\dfrac{2}{4}=\dfrac{1}{2}$

$\Rightarrow |z|^2=0.50$

Question: Let $\bar{z}$ denote the complex conjugate of a complex number z and let $i=\sqrt{-1}$. In the set of complex numbers, the number of distinct roots of the equation $\bar{z}-z^2=\left(\bar{z}+z^2\right)$ is ____.

Answer: 4.00

Solution:

Given,

$\bar{z}-z^2=i\left(\bar{z}+z^2\right)$

$\Rightarrow (1-i)\bar{z}=(1+i)z^2$

$\Rightarrow \dfrac{(1-i)}{(1+i)}\bar{z}=z^2$

$\Rightarrow \left(-\dfrac{2i}{2}\right)\bar{z}=z^2$

$\therefore z^2=-i\bar{z}$

Let $z=x+iy$,

$\therefore \left(x^2-y^2\right)+i(2xy)=-i(x-iy)$

So, $x^2-y^2+y=0 \qquad ....(1)$

and $(2y+1)x=0 \qquad ....(2)$

$\Rightarrow x=0$ or $y=-\dfrac{1}{2}$

Case I: when $x=0$

$\therefore (1)\Rightarrow y(1-y)=0\Rightarrow y=0,1$

$\therefore (0,0),(0,1)$

Case II: when $y=-\dfrac{1}{2}$

$\therefore (1)\Rightarrow x^2-\dfrac{1}{4}-\dfrac{1}{2}=0\Rightarrow x^2=\dfrac{3}{4}\Rightarrow x=\pm\dfrac{\sqrt{3}}{2}$

$\therefore \left(\dfrac{\sqrt{3}}{2},-\dfrac{1}{2}\right),\left(-\dfrac{\sqrt{3}}{2},-\dfrac{1}{2}\right)$

$\Rightarrow$ Number of distinct 'z' is equal to 4.

Question: Let $l_1, l_2, \ldots, l_{100}$ be consecutive terms of an arithmetic progression with common difference d_1, and let $w_1, w_2, \ldots, w_{100}$ be consecutive terms of another arithmetic progression with common difference d_2 where $d_1 d_2 = 10$. For each $i = 1, 2, \ldots., 100$, let R_i be rectangle with length l_i, width w_i and area A_i. If $A_{51} - A_{50} = 1000$, then the value of $A_{100} - A_{90}$ is ____.

Answer: 18900.00

Solution:

Given, $A_{51} - A_{50} = 1000 \Rightarrow l_{51}w_{51} - l_{50}w_{50} = 1000$

$\Rightarrow (l_1 + 50d_1)(w_1 + 50d_2) - (l_1 + 49d_1)(w_1 + 49d_2) = 1000$

$\Rightarrow (l_1 d_2 + w_1 d_1) = 10 \qquad \ldots.(1)$

(As $d_1 d_2 = 10$)

$\therefore A_{100} - A_{90} = l_{100}w_{100} - l_{90}w_{90}$

$= (l_1 + 99d_1)(w_1 + 99d_2) - (l_1 + 89d_1)(w_1 + 99d_2)$

$= 10(l_1 d_2 + w_1 d_1) + (99^2 - 89^2) d_1 d_2$

$= 10(10) + \underbrace{(99 - 89)}_{=10}(99 + 89)(10)$

(As, $d_1 d_2 = 10$)

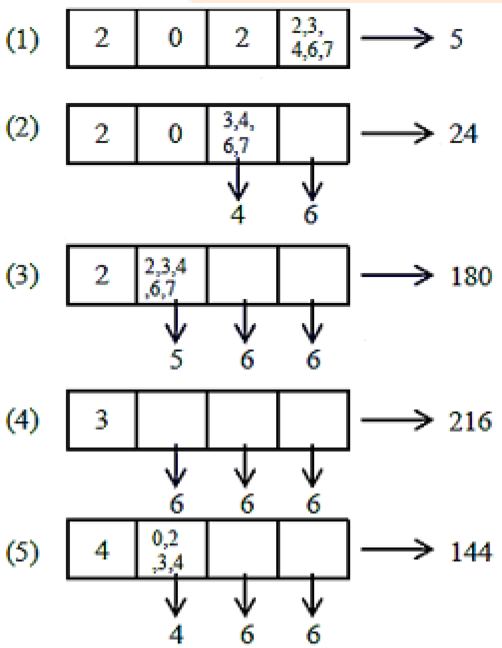
$= 100(1 + 188) = 100(189)$

$= 18900$

Question: The number of 4-digit integers in the closed interval $[2022, 4482]$ formed by using the digits 0, 2, 3, 4, 6, 7 is ____________.

Answer: 569.00

Solution:



Number of 4 digit integers in $[2022, 4482]$

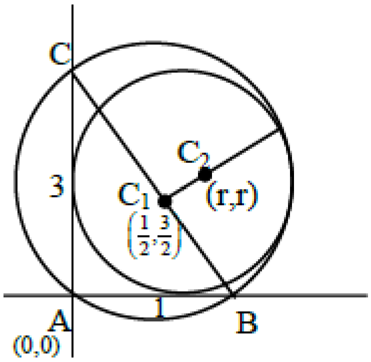
$= 5 + 24 + 180 + 216 + 144 = 569$

Question: Let ABC be the triangle with $AB = 1$, $AC = 3$ and $\angle BAC = \frac{\pi}{2}$. If a circle of radius $r > 0$ touches the sides AB, AC and also touches internally the circumcircle of the triangle ABC, then the value of r is ____.

Answer: 0.83 or 0.84

Solution:

$4 - \sqrt{10} = 0.83$ or 0.84



$C_1\left(\frac{1}{2},\frac{3}{2}\right)$ and $r_1 = \frac{\sqrt{10}}{2}$

$C_2 = (r, r)$

$\therefore$ Circle C_2 touches C_1 internally

$$\Rightarrow C_1C_2 = \left| r - \frac{\sqrt{10}}{2} \right|$$

$$\Rightarrow \left(r - \frac{1}{2}\right)^2 + \left(r - \frac{3}{2}\right)^2 = \left(r - \frac{\sqrt{10}}{2}\right)^2$$

$r^2 - 4r + \sqrt{10}r = 0$

$r = 0$ (reject) or $r = 4 - \sqrt{10}$

Question: Consider the equation

$$\int_1^e \frac{(\log x)^{\frac{1}{2}}}{x\left(a - (\log_e x)^{\frac{3}{2}}\right)^2} dx = 1,\ a \in (-\infty, 0) \cup (1, \infty).$$

Which of the following statements is/are TRUE ?

Options:

(a) No a satisfies the above equation

(b) An integer a satisfies the above equation

(c) An irrational number a satisfies the above equation

(d) More than one a satisfy the above equation

Answer: (c), (d)

Solution:

$$\int_1^e \frac{(\log_e x)^{\frac{1}{2}}}{x\left(a-(\log_e x)^{\frac{3}{2}}\right)^2}=1$$

Let $a-(\log_e x)^{\frac{3}{2}}=t$

$$\frac{(\log_e x)^{\frac{1}{2}}}{x}dx=-\frac{2}{3}t$$

$$=\frac{2}{3}\int_a^{a-1}\frac{-dt}{t^2}=\frac{2}{3}\left(\frac{1}{t}\right)_a^{a-1}=1$$

$$\frac{2}{3a(a-1)}=1$$

$$3a^2-3a-2=0$$

$$a=\frac{3\pm\sqrt{33}}{6}$$

Question: Let $a_1, a_2, a_3, \ldots$ be an arithmetic progression with $a_1=7$ and common difference 8. Let $T_1, T_2, T_3, \ldots.$ be such that $T_1=3$ and $T_{n+1}-T_n=a_n$ for $n\geq 1$. Then, which of the following is/are TRUE?

Options:

(a) $T_{20}=1604$

(b) $\sum_{k=1}^{20} T_k=10510$

(c) $T_{30}=3454$

(d) $\sum_{k=1}^{30} T_k=35610$

Answer: (b), (c)

Solution:

$a_1=7,\ d=8$

$T_{n+1}-T_n=a_n \forall\ n\geq 1$

$S_n=T_1+T_2+T_3+\ldots.+T_{n-1}+T_n$

$S_n=\quad T_1+T_2+T_3+\ldots.+T_{n-1}+T_n$

on subtraction

$T_n=T_1+a_1+a_2+\ldots.+a_{n-1}$

$T_n=3+(n-1)(4n-1)$

$T_n=4n^2-5n+4$

$$\sum_{k=1}^{n} T_k=4\sum n^2-5\sum n+4n$$

$T_{20}=1504$

$T_{30}=3454$

$\sum_{k=1}^{30} T_k = 35615$

$\sum_{k=1}^{20} T_k = 10510$

Question: Let P_1 and P_2 be two planes given by $P_1 : 10x + 15y + 12z - 60 = 0,\ P_2 : -2x + 5y + 4z - 20 = 0.$ Which of the following straight lines can be an edge of some tetrahedron whose two faces lie on P_1 and P_2

Options:

(a) $\frac{x-1}{0} = \frac{y-1}{0} = \frac{z-1}{5}$

(b) $\frac{x-6}{-5} = \frac{y}{2} = \frac{z}{3}$

(c) $\frac{x}{-2} = \frac{y-4}{5} = \frac{z}{4}$

(d) $\frac{x}{1} = \frac{y-4}{-2} = \frac{z}{3}$

Answer: (a), (b), (d)

Solution:

Line of intersection is $\frac{x}{0} = \frac{y-4}{-4} = \frac{z}{5}$

(1) Any skew line with the line of intersection of given planes can be edge of tetrahedron.

(2) any intersecting line with line of intersection of given planes must lie either in plane P_1 or P_2 can be edge of tetrahedron.

Question: Let S be the reflection of a point Q with respect to the plane given by $\vec{r} = -(t+p)\hat{i} + t\hat{j} + (1+p)\hat{k}$ where t, p are real parameters and $\hat{i}, \hat{j}, \hat{k}$ are the unit vectors along the three positive coordinate axes. If the position vectors of Q and S are $10\hat{i} + 15\hat{j} + 20\hat{k}$ and $\alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$ respectively, then which of the following is/are TRUE ?

Options:

(a) $3(\alpha + \beta) = -101$

(b) $3(\beta + \gamma) = -71$

(c) $3(\gamma + \alpha) = -86$

(d) $3(\alpha + \beta + \gamma) = -121$

Answer: (a), (b), (c)

Solution:

$\vec{r} = \hat{k} + t(-\hat{i} + \hat{j}) + p(-\hat{i} + \hat{k})$

$\vec{n} = \hat{i} + \hat{j} + \hat{k}$

$\Rightarrow x + y + z = 1$

$Q(10,15,20)$ and $S(\alpha, \beta, \gamma)$

$$\frac{\alpha-10}{1}=\frac{\beta-15}{1}=\frac{\gamma-20}{1}=-2\left(\frac{10+15+20-1}{1+1+1}\right)$$
$$=-\frac{88}{3}$$
$$\Rightarrow(\alpha,\beta,\gamma)\equiv\left(-\frac{58}{3},-\frac{43}{3},-\frac{28}{3}\right)$$
$\Rightarrow A,B,C$ are correct options

Question: Consider the parabola $y^2=4x$. Let S be the focus of the parabola. A pair of tangents drawn to the parabola from the point $P=(-2,1)$ meet the parabola at P_1 and P_2. Let Q_1 and Q_2 be points on the lines SP_1 and SP_2 respectively such that PQ_1 is perpendicular to SP_1 and PQ_2 is perpendicular to SP_2. Then, which of the following is/are TRUE ?

Options:

(a) $SQ_1=2$

(b) $Q_1Q_2=\frac{3\sqrt{10}}{5}$

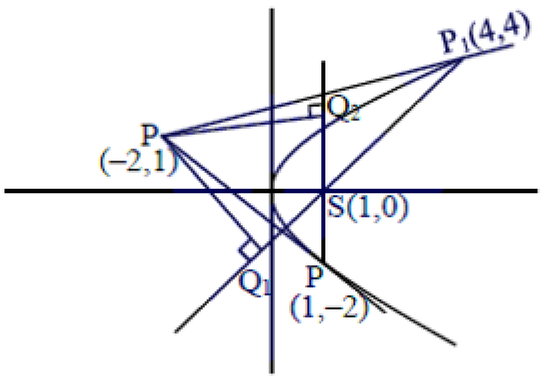
(c) $PQ_1=3$

(d) $SQ_2=1$

Answer: (b), (c), (d)

Solution:

Let equation of tangent with slope ‘m’ be



$$T: y=mx+\frac{1}{m}$$
$T:$ passes through (-2, 1) so
$$1=-2m+\frac{1}{m}$$
$$\Rightarrow m=-1 \text{ or } m=\frac{1}{2}$$
Points are given by $\left(\frac{a}{m^2},\frac{2a}{m}\right)$

So, one point will be (1, –2) & (4, 4)

Let $P_1(4,4)$ & $P_2(1,-2)$

$P_1S : 4x-3y-4=0$

$P_2S = x-1=0$

$$PQ_1 = \left|\frac{4(-2)-3(1)-4}{5}\right| = 3$$

$SP=\sqrt{10};\ PQ_2=3;\ SQ_1=1=SQ_2$

$$\frac{1}{2}\left(\frac{Q_1Q_2}{2}\right)\times\sqrt{10}=\frac{1}{2}\times 3\times 1 \text{ (comparing Areas)}$$

$$\Rightarrow Q_1Q_2=\frac{2\times 3}{\sqrt{10}}=\frac{3\sqrt{10}}{5}$$

Question: Let $|M|$ denote the determinant of a square matrix M. Let $g:\left[0,\frac{\pi}{2}\right]\to\mathbb{R}$ be the function defined by $g(\theta)=\sqrt{f(\theta)-1}+\sqrt{f\left(\frac{\pi}{2}-\theta\right)-1}$ where

$$f(\theta)=\frac{1}{2}\begin{vmatrix} 1 & \sin\theta & 1 \\ -\sin\theta & 1 & \sin\theta \\ -1 & -\sin\theta & 1 \end{vmatrix}+\begin{vmatrix} \sin\pi & \cos\left(\theta+\frac{\pi}{4}\right) & \tan\left(\theta-\frac{\pi}{4}\right) \\ \sin\left(\theta-\frac{\pi}{4}\right) & -\cos\frac{\pi}{2} & \log_e\left(\frac{4}{\pi}\right) \\ \cot\left(\theta+\frac{\pi}{4}\right) & \log_e\left(\frac{\pi}{4}\right) & \tan\pi \end{vmatrix}$$

Let (x) be a quadratic polynomial whose roots are the maximum and minimum values of the function (θ), and $(2)=2-\sqrt{2}$. Then which of the following is/are TRUE ?

Options:

(a) $p\left(\frac{3+\sqrt{2}}{4}\right)<0$

(b) $p\left(\frac{1+3\sqrt{2}}{4}\right)>0$

(c) $p\left(\frac{5\sqrt{2}-1}{4}\right)>0$

(d) $p\left(\frac{5-\sqrt{2}}{4}\right)<0$

Answer: (a), (c)

Solution:

$$f(\theta)=\frac{1}{2}\begin{vmatrix}1 & \sin\theta & 1\\ -\sin\theta & 1 & \sin\theta\\ -1 & -\sin\theta & 1\end{vmatrix}+\begin{vmatrix}\sin\pi & \cos\left(\theta+\frac{\pi}{4}\right) & \tan\left(\theta-\frac{\pi}{4}\right)\\ \sin\left(\theta-\frac{\pi}{4}\right) & -\cos\frac{\pi}{2} & \log_e\left(\frac{4}{\pi}\right)\\ \cot\left(\theta+\frac{\pi}{4}\right) & \log_e\left(\frac{\pi}{4}\right) & \tan\pi\end{vmatrix}$$

$$f(\theta)=\frac{1}{2}\begin{vmatrix}2 & \sin\theta & 1\\ 0 & 1 & \sin\theta\\ 0 & -\sin\theta & 1\end{vmatrix}+\begin{vmatrix}0 & -\sin\left(\theta-\frac{\pi}{4}\right) & \tan\left(\theta-\frac{\pi}{4}\right)\\ \sin\left(\theta-\frac{\pi}{4}\right) & 0 & \log_e\left(\frac{4}{\pi}\right)\\ -\tan\left(\theta-\frac{\pi}{4}\right) & -\log_e\left(\frac{4}{\pi}\right) & 0\end{vmatrix}$$

$f(\theta)=\left(1+\sin^2\theta\right)+0$ (skew symmetric)

$g(\theta)=\sqrt{f(\theta)-1}+\sqrt{f\left(\frac{\pi}{2}-\theta\right)-1}$

$=|\sin\theta|+|\cos\theta|$ for $\theta\in\left[0,\frac{\pi}{2}\right]$

$g(\theta)\in\left[1,\sqrt{2}\right]$

Again let $P(x)=k\left(x-\sqrt{2}\right)(x-1)$

$2-\sqrt{2}=k\left(2-\sqrt{2}\right)(2-1)$

$\Rightarrow k=1$ ($P(2)=2-\sqrt{2}$ given)

$\therefore P(x)=\left(x-\sqrt{2}\right)(x-1)$

For option (A) $P\left(\frac{3+\sqrt{2}}{4}\right)<0$ correct

Option (B) $P\left(\frac{1+3\sqrt{2}}{4}\right)<0$ incorrect

Option (C) $P\left(\frac{5\sqrt{2}-1}{4}\right)>0$ correct

Option (D) $P\left(\frac{5-\sqrt{2}}{4}\right)>0$ incorrect

Question: Consider the following lists:

List-I	**List-II**
(I) $\left\{x\in\left[-\frac{2\pi}{3},\frac{2\pi}{3}\right]:\cos x+\sin x=1\right\}$	(P) has two elements
(II) $\left\{x\in\left[-\frac{5\pi}{18},\frac{5\pi}{18}\right]:\sqrt{3}\tan 3x=1\right\}$	(Q) has three elements

(III) $\left\{x \in \left[-\frac{6\pi}{5}, \frac{6\pi}{5}\right] : 2\cos(2x) = \sqrt{3}\right\}$	(R) has four elements
(IV) $\left\{x \in \left[-\frac{7\pi}{4}, \frac{7\pi}{4}\right] : \sin x - \cos x = 1\right\}$	(S) has five elements
	(T) has six elements

The correct option is:

Options:

(a) (I) → (P); (II) → (S); (III) → (P); (IV) → (S)
(b) (I) → (P); (II) → (P); (III) → (T); (IV) → (R)
(c) (I) → (Q); (II) → (P); (III) → (T); (IV) → (S)
(d) (I) → (Q); (II) → (S); (III) → (P); (IV) → (R)

Answer: (b)

Solution:

(I) $\left\{x \in \left[-\frac{2\pi}{3}, \frac{2\pi}{3}\right] : \cos x + \sin x = 1\right\}$

$\cos x + \sin x = 1$

$\Rightarrow \frac{1}{\sqrt{2}}\cos x + \frac{1}{\sqrt{2}}\sin x = \frac{1}{\sqrt{2}} \Rightarrow \cos\left(x - \frac{\pi}{4}\right) = \cos\frac{\pi}{4}$

$\Rightarrow x - \frac{\pi}{4} = 2n\pi \pm \frac{\pi}{4}; n \in Z \Rightarrow x = 2n\pi; x = 2n\pi + \frac{\pi}{2};\ nkeZ$

$\Rightarrow x \in \left\{0, \frac{\pi}{2}\right\}$ in given range has two solutions

(II) $\left\{x \in \left[-\frac{5\pi}{18}, \frac{5\pi}{18}\right] : \sqrt{3}\tan 3x = 1\right\}$

$\sqrt{3}\tan 3x = 1 \Rightarrow \tan 3x = \frac{1}{\sqrt{3}} \Rightarrow 3x = n\pi + \frac{\pi}{6}$

$\Rightarrow x = (6n+1)\frac{\pi}{18};\ n \in Z$

$\Rightarrow x \in \left\{\frac{\pi}{18}, \frac{-5\pi}{18}\right\}$ in given range has two solutions

(III) $\left\{x \in \left[-\frac{6\pi}{5}, \frac{6\pi}{5}\right] : 2\cos(2x) = \sqrt{3}\right\}$

$2\cos 2x = \sqrt{3}$

$\Rightarrow \cos 2x = \frac{\sqrt{3}}{2} = \cos\frac{\pi}{6}$

$\Rightarrow 2x = 2n\pi \pm \frac{\pi}{6};\ n \in Z$

$\Rightarrow x = n\pi \pm \frac{\pi}{12};\ n \in Z$

$x \in \left\{\pm\frac{\pi}{12}, \pi \pm \frac{\pi}{12}, -\pi \pm \frac{\pi}{12}\right\}$

Six solutions in given range

(IV) $\left\{x \in \left[-\frac{7\pi}{4}, \frac{7\pi}{4}\right] : \sin x - \cos x = 1\right\}$

$\cos x - \sin x = -1$

$\Rightarrow \cos\left(x + \frac{\pi}{4}\right) = \frac{-1}{\sqrt{2}} = \cos\frac{3\pi}{4}$

$\Rightarrow x + \frac{\pi}{4} = 2n\pi \pm \frac{3\pi}{4}; n \in Z$

$\Rightarrow x = 2n\pi + \frac{\pi}{2}$ or $x = 2n\pi - \pi;\ n \in Z$

$\Rightarrow x \in \left\{\frac{\pi}{2}, \frac{-3\pi}{2}, \pi, -\pi\right\}$ four solutions in given range

Question: Two players P_1 and P_2, play a game against each other. In every round of the game, each player rolls a fair die once, where the six faces of the die have six distinct numbers. Let x and y denote the readings on the die rolled by P_1 and P_2, respectively. If $x > y$, then P_1 scores 5 points and P_2 scores 0 point. If $x = y$, then each player scores 2 points. If $x < y$, then P_1 scores 0 point and P_2 scores 5 points. Let X_i and Y_i be the total scores of P_1 and P_2, respectively, after playing the i^{th} round.

List-I	**List-II**
(I) Probability of $(X_2 \geq Y_2)$ is	(P) $\frac{3}{8}$
(II) Probability of $(X_2 > Y_2)$ is	(Q) $\frac{11}{16}$
(III) Probability of $(X_3 = Y_3)$ is	(R) $\frac{5}{16}$
(IV) Probability of $(X_3 > Y_3)$ is	(S) $\frac{355}{864}$
	(T) $\frac{77}{432}$

Options:
(a) (I) → (Q); (II) → (R); (III) → (T); (IV) → (S)
(b) (I) → (Q); (II) → (R); (III) → (T); (IV) → (T)
(c) (I) → (P); (II) → (R); (III) → (Q); (IV) → (S)
(d) (I) → (P); (II) → (R); (III) → (Q); (IV) → (T)
Answer: (a)
Solution:

P(draw in 1 round) $= \frac{6}{36} = \frac{1}{6}$

P(win in 1 round) $= \frac{1}{2}\left(1 - \frac{1}{6}\right) = \frac{5}{12}$

P(loss in 1 round) $= \frac{5}{12}$

$$P(X_2 > Y_2) = P(10,0) + P(7,2) = \frac{5}{12}\times\frac{5}{12} + \frac{5}{12}\times\frac{1}{6}\times 2 = \frac{45}{144} = \frac{5}{16}$$

$$P(X_2 = Y_2) = P(5,5) + P(4,4) = \frac{5}{12}\times\frac{5}{12}\times 2 + \frac{1}{6}\times\frac{1}{6} = \frac{25+2}{72} = \frac{3}{8}$$

$$P(X_3 = Y_3) = P(6,6) + P(7,7) = \frac{1}{6\times 6\times 6} + \frac{5}{12}\times\frac{1}{6}\times\frac{5}{12}\times 6 = \frac{2}{432} + \frac{75}{432} = \frac{77}{432}$$

$$P(X_3 > Y_3) = \frac{1}{2}\left(1 - \frac{77}{432}\right) = \frac{355}{864}$$

Question: Let p, q, r be nonzero real numbers that are, respectively, the 10th, 100th and 1000th terms of a harmonic progression. Consider the system of linear equations

$x + y + z = 1$

$10x + 100y + 1000z = 0$

$qr\,x + pr\,y + pq\,z = 0$

List-I	**List-II**
(I) If $\frac{q}{r} = 10,$ then the system of linear equation has	(P) $x = 0, y = \frac{10}{9}, z = -\frac{1}{9}$ as a solution
(II) If $\frac{p}{r} \neq 100,$ then the system of linear equation has	(Q) $x = \frac{10}{9}, y = -\frac{1}{9}, z = 0$ as a solution
(III) If $\frac{p}{q} \neq 10,$ then the system of linear equation has	(R) infinitely many solutions
(IV) If $\frac{p}{q} = 10,$ then the system of linear equation has	(S) no solution
	(T) at least one solution

The correct option is:

Options:

(a) (I) → (T); (II) → (R); (III) → (S); (IV) → (T)

(b) (I) → (Q); (II) → (S); (III) → (S); (IV) → (R)

(c) (I) → (Q); (II) → (R); (III) → (P); (IV) → (R)

(d) (I) → (T); (II) → (S); (III) → (P); (IV) → (T)

Answer: (b)

Solution:

If $\frac{q}{r} = 10 \Rightarrow A = D \Rightarrow D_x = D_y = D_z = 0$

So, there are infinitely many solutions

Look of infinitely many solutions can be given as

$x + y + z = 1$

& $10x + 100y + 1000z = 0 \Rightarrow x + 10y + 100z = 0$

Let $z = \lambda$

Then $x+y=1-\lambda$

And $x+10y=-100\lambda$

$\Rightarrow x=\frac{10}{9}+10\lambda;\ y=\frac{-1}{9}-11\lambda$

i.e., $(x,y,z)\equiv\left(\frac{10}{9}+10\lambda,\frac{-1}{9}-11\lambda,\lambda\right)$

$Q\left(\frac{10}{9},\frac{-1}{9},0\right)$ valid for $\lambda=0$

$P\left(0,\frac{10}{9},\frac{-1}{9}\right)$ not valid for any λ.

(I) $\rightarrow$ Q, R, T

(II) If $\frac{p}{r}\neq 100$, then $D_y\neq 0$

So no solution

(II) $\rightarrow$ (S)

(III) If $\frac{p}{q}\neq 10$, then $D_z\neq 0$ so, no solution

(III) $\rightarrow$ (S)

(IV) If $\frac{p}{q}=10\Rightarrow D_z=0\Rightarrow D_x=D_y=0$

So infinitely many solution

(IV) $\rightarrow$ Q, R, T

Question: Consider the ellipse $\frac{x^2}{4}+\frac{y^2}{3}=1$. Let $(\alpha,0), 0<\alpha<2$, be a point. A straight line drawn through H parallel to the y-axis crosses the ellipse and its auxiliary circle at points E and F respectively, in the first quadrant. The tangent to the ellipse at the point E intersects the positive x-axis at a point G. Suppose the straight line joining F and the origin makes an angle ϕ with the positive x-axis.

List-I	**List-II**
(I) If $\phi=\frac{\pi}{4}$, then the area of the triangle FGH is	(P) $\frac{(\sqrt{3}-1)^4}{8}$
(II) If $\phi=\frac{\pi}{3}$, then the area of the triangle FGH is	(Q) 1
(III) If $\phi=\frac{\pi}{6}$, then the area of the triangle FGH is	(R) $\frac{3}{4}$
(IV) If $\phi=\frac{\pi}{12}$, then the area of the triangle FGH is	(S) $\frac{1}{2\sqrt{3}}$
	(T) $\frac{3\sqrt{3}}{2}$

The correct option is:

Options:

(a) (I) → (R); (II) → (S); (III) → (Q); (IV) → (P)

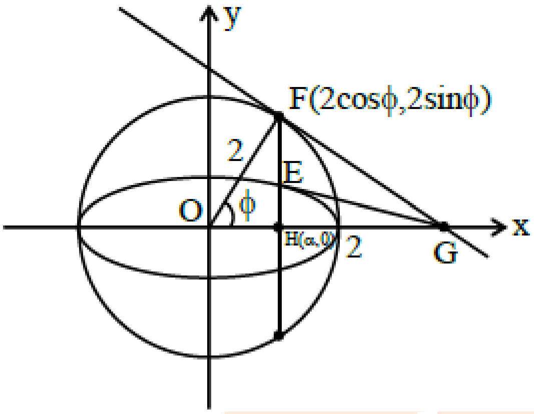
(b) (I) → (R); (II) → (T); (III) → (S); (IV) → (P)

(c) (I) → (Q); (II) → (T); (III) → (S); (IV) → (P)

(d) (I) → (Q); (II) → (S); (III) → (Q); (IV) → (P)

Answer: (c)

Solution:



Let $F(2\cos\phi, 2\sin\phi)$ & $E(2\cos\phi, \sqrt{3}\sin\phi)$

$$EG: \frac{x}{2}\cos\phi + \frac{y}{\sqrt{3}}\sin\phi = 1$$

$$\therefore\ G\left(\frac{2}{\cos\phi}, 0\right) \text{ and } \alpha = 2\cos\phi$$

$$ar(\Delta FGH) = \frac{1}{2} HG \times FH$$

$$= \frac{1}{2}\left(\frac{2}{\cos\phi} - 2\cos\phi\right) \times 2\sin\phi$$

$$f(\phi) = 2\tan\phi\sin^2\phi$$

$$\therefore\ \text{(I)}\ f\left(\frac{\pi}{4}\right) = 1$$

$$\text{(II)}\ f\left(\frac{\pi}{3}\right) = \frac{3\sqrt{3}}{2}$$

$$\text{(III)}\ f\left(\frac{\pi}{6}\right) = \frac{1}{2\sqrt{3}}$$

$$\text{(IV)}\ f\left(\frac{\pi}{12}\right) = 2\left(2-\sqrt{3}\right)\left(\frac{\sqrt{3}-1}{2\sqrt{2}}\right)^2 = \left(4-2\sqrt{3}\right)\frac{\left(\sqrt{3}-1\right)^2}{8} = \frac{\left(\sqrt{3}-1\right)^4}{8}$$

∴ (I) → (Q); (II) → (T); (III) → (S); (IV) → (P)